Minimum Extendable Fano Projections for Five-Cycle Double Covers: Exact Frontiers and Failure of Minimum Selection

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Resolution status. The five-cycle double cover conjecture remains unresolved. This note proves an exchange theorem, settles the minimum-support cases through size fifteen for finite connected bridgeless loopless cubic multigraphs (parallel edges allowed), and gives exact finite censuses. It also completes a one-round matching-game analysis on the eight residual states in one fixed support-sixteen word orbit, without settling the unrestricted support-sixteen case. It also constructs a 162-vertex graph whose unique minimum extendable projection is not cleanable, refuting the proposed minimum-selection route. That graph has an explicit five-cycle double cover. Thus the note proves neither FiveCDC nor its negation. The standard conjecture for finite bridgeless multigraphs reduces to the cubic setting by Proposition 1.1. This status is distinct from the ordinary cycle double cover theorem proved in July 2026.

Abstract

Hušek and Šámal recently expressed the five-cycle double cover conjecture as the problem of selecting a nowhere-zero $\mathbb{F}_2^3$-flow with a component-parity property. We study the support h of one binary coordinate of such a flow. Call h *extendable* if it occurs as a coordinate of some nowhere-zero $\mathbb{F}_2^3$-flow.

We prove a simultaneous exchange theorem for a minimum-cardinality extendable projection. Write the flow as $f = (h, s)$, with $s : E \to \mathbb{F}_2^2$, and partition h into the four affine value classes $M_c = \{e \in h : s(e) = c\}$. For every c, the projection h is a minimum-cardinality binary cycle containing M_c. Equivalently, every binary cycle C avoiding M_c satisfies

$$|C \cap h| \leq |C - h|.$$

Thus every circuit component of h contains all four affine colours, and every support-shortening cycle meets all four classes. The proof is the addition of the Fano value $(1, c)$ along C.

We first prove, without computation, that every globally minimum extendable projection of size at most five is cleanable. We then give a stronger direct boundary repair. Independently on every component of $G - h$, apply an element of $\mathrm{GL}(2, 2)$ to the low-flow values; on every circuit of h, solve the resulting cyclic vertex equations while allowing zero low values, since the first coordinate is one there. Cleanliness becomes four explicit cut-parity equations per component. An exact affine/dihedral quotient exhausts every support shape, proper four-colour circuit word, and component partition through total support size ten. All 125,178 valid dirty canonical states admit a direct repair. At size eleven the same test first fails: there are 14 failed $5 + 6$ boundary states, in 11 full symmetry orbits. For every failure, however, an explicit boundary certificate makes the six-circuit low-valued and nowhere zero, so deleting it from the

first-coordinate support produces an extendable projection of size five. Global minimality excludes all 14. Thus every minimum extendable projection of size at most eleven is cleanable. At size twelve, 13,788,824 dirty boundary states either clean directly or admit a strict circuit deletion; two independent enumerators agree exactly. Thus every minimum extendable projection of size at most twelve is cleanable. At size thirteen two independently written exact classifiers agree on all five support shapes and all 189,998,862 charge-valid states. Of the 159,369,966 dirty states, 159,362,292 clean directly and 7,674 admit a strict circuit deletion; there are zero residuals. Thus every minimum extendable projection of size at most thirteen is cleanable.

At size fourteen the primary exact classifier enumerates 9,481 word orbits and 2,616,134,989 charge-valid states. Of the 2,255,478,176 dirty states, 2,255,331,588 clean directly and 146,364 admit a strict circuit deletion. The 224 residuals all have shape $7+7$. A separately written full $7+7$ enumerator independently reproduces all 333 word orbits, 91,481,505 charge-valid states, and the identical residual set. We prove a two-colour path-switch lemma for the split-occurrence boundary model and give 724 literal deletion rows that meet every possible four- or six-terminal path pairing. Every residual therefore has a strict size-seven deletion escape in every cubic realization. Thus every minimum extendable projection of size at most fourteen is cleanable.

We next prove a universal relative-$\mathrm{GL}(2,2)$ tensor lemma. Given an arbitrary linear functional of $L_a^{-1}L_b$ on every pair of complement components, component maps can always be chosen so that the resulting pair graph is Eulerian. A short human proof uses a three-point linear functional on the six elements of $\mathrm{GL}(2,2)$; after expanding an indicator polynomial, leaf terms vanish and nonempty cycle-union terms cancel in pairs. This implies that every charge-valid one-support-circuit boundary state is directly cleanable, with no bound on the circuit length or number of complement components. It also cleans several support circuits whenever every component is charge-balanced separately on each circuit. The latter hypothesis is sharp for direct cleaning: a two-occurrence interaction-flow dictionary reduces the first abstract failure to the Petersen graph. Its displayed two-5-circuit support has no clean extension, but every extension strictly deletes one circuit and reaches a clean global minimum of size five.

For general several-circuit states we reduce fixed-map cleanliness to an affine XOR system with two binary variables per relative circuit translation. Every inconsistency has a dual component-union witness. Postcomposing precisely those component maps by one common element of $\mathrm{GL}(2,2)$ preserves integrability. A new six-map parity identity proves that one of the six choices always neutralizes that particular witness. The proof is a four-by-four quadratic table followed by cancellation on cyclic selected and unselected runs. This is single-witness neutralization, not simultaneous cleaning: the sharp size-fourteen state shows that another witness can remain or appear. Indeed, its forty reduced map states are strongly connected under legal neutralizations and contain a displayed four-step cycle. A human-visible exchange cycle excludes that orbit from global minimality in its 18-vertex realization. Thus any convergence theorem must use global-minimum data essentially.

We make that requirement precise in the four shortest-join duals. Every dual cut which is a union of components of $G-h$ has coefficient zero in every optimal dual; all positive dual shores split the interior of a complement component. The rainbow witness cut is therefore already zero-priced before neutralization, and the aggregate four-dual optimum is always $3|h|$. Double-counting the forced interior dual load gives the new density bounds

$$|E-h| \geq |h| - |M_c| \quad (c \in K), \qquad |h| \leq \tfrac{6}{7}|V|$$

for a cubic graph. Hence an LP descent must transport metric or path-pairing information inside complement components, not merely a quotient-cut price. An exact slack decomposition shows that equality $|h| = 6|V|/7$ is rigid: $G-h$ is a forest, all complement and boundary derivative colours are equinumerous, $|h|$ is divisible by twelve, and every affine witness is exactly colour-balanced. Any uncleanable equality obstruction has $|h| \geq 24$.

We also obtain an exact dynamic exchange test. Every nonzero functional α on the low $\mathbb{F}_2^2$-flow takes value one on at least $|h|/2$ complement edges. If Y is a six-map-integrable union of complement components, and N_b, O_a count complement edges of colours $b, a \neq 0$ inside and outside Y, respectively, then

$$O_a + N_b \le |E - h| - \tfrac{1}{2}|h| \qquad (a, b \ne 0).$$

Violation explicitly returns a legal common map switch and a strict colour-avoiding binary-cycle descent. In boundary form this is $k_a(\overline{Y}) + k_b(Y) \le 2(|V| - |h|)$. The unresolved implication is now concrete: prove that some rainbow-odd witness violates this load bound, or use the interior geometry that lets all nine inequalities survive.

The first alternative is false from local boundary data alone. On the cube graph, a displayed 6-circuit projection and its two complementary 3-stars have a rainbow-odd, six-map-integrable witness, use all four support colours, and satisfy every colour-load inequality strictly. Lengths four and five are impossible in the corresponding two-tree model, so six is the first local counterstate. The cube is Tait-colourable and four of its six map switches already clean the state; the projection is not globally minimum. Thus the counterstate isolates, rather than bypasses, the missing global hypothesis. A stronger support-fourteen survivor has no clean or deletion outcome under any component-map tuple, satisfies all nine load inequalities, and has a 278-vertex realization attaining all four complete static shortest-join optima. Its fixed projection is uncleanable, but its exact global minimum is seven. Hence the known static consequences of minimality are collectively insufficient. This initially points toward dynamic use of the actual minimum; the strict-lock construction below shows that even full global minimality does not force cleaning.

We therefore give the exact global optimization, not only its four affine branches. Choose any low $\mathbb{F}_2^2$-flow s', let M be its exact zero set, and choose $J \subseteq E - M$ with $\partial J = \partial M$. Every extendable projection, uniquely in this branch, is $M \mathbin{\dot\cup} J$, and hence

$$\mu(G) = \min_{s', J}\bigl(|M| + |J|\bigr).$$

For a feasible cubic branch M is a matching and the inner problem is an ordinary shortest-∂M-join problem. Cleanliness is exactly the existence of a second ∂M-join disjoint from $M \cup J$. Two independent checkers reconstruct the 278-vertex survivor and its coupled support-seven competitor.

The resulting minimum-selection principle is nevertheless false. A strict parity-lock two-pole has conditional projection minima $(a_0, a_1) = (7, 5)$. Substituting eight copies into an 18-vertex dirty base gives a simple connected bridgeless nonplanar cubic graph on 162 vertices and 243 edges. Its unique globally minimum projection has size 54, is the union of two 27-circuits, and has no clean extension. The mandatory FiveCDC exact-two/XOR formula is SAT on this graph, and a short compositional proof glues five-covers of the base and lock closure. Thus the construction refutes minimum selection as a proof strategy but is positively not a FiveCDC counterexample.

At size fifteen the primary exact classifier enumerates 26,492 word orbits and 35,247,202,556 charge-valid abstract states over all eight support shapes. Of the 31,088,622,592 dirty states, 31,086,255,789 clean directly and 2,360,767 admit a strict circuit deletion. The remaining 6,036 states all have shape $7 + 8$. A separately written canonical audit identifies every one as a same-block one-occurrence extension of a frozen size-fourteen residual, and realization-robust inverse Kempe certificates cover all of them. The 20,004-word one-circuit row independently agrees with a zero-sum-partition recurrence; all 23,398,774,592 dirty states also clean in the completed anchor census, in agreement with the tensor proof. Therefore every minimum extendable projection of size at most fifteen is cleanable, and a counterexample to the proposed selection principle must have minimum size at least sixteen.

In the two-occurrence interaction subclass we close every state through support sixteen, including interaction loops, and every loopless state through support eighteen. The loopless support-eighteen census has 1,041,984 cyclic-order states: 1,035,036 clean and 6,948 delete-only. A separate human shape reduction leaves four loop profiles through support sixteen; an exact 16,344-state census has zero residuals. Thus every globally minimum state in these subclasses is cleanable. Arbitrary occurrence multiplicities remain open already at size sixteen.

For one fixed $8+8$ support-word orbit at size sixteen, an exact canonical census goes beyond the two-occurrence subclass. Of 8,046,330 charge-valid component partitions, 8,041,808 clean directly, 4,514 delete a circuit, and eight are residual. In a one-colour-pair matching game, six of those eight admit a $\{1,3\}$-Kempe strategy after observing every abstract terminal matching. The two $8+2+2+2+2$ residuals admit jointly realizable adverse matchings for all three colour pairs; two explicit 42-vertex simple bridgeless cubic realizations are Tait-colourable and defeat every legal one-round move in the stated game. This is a theorem about one word orbit and one-round support-preserving play. The displayed projections are not globally minimum, and the result is neither a FiveCDC counterexample nor a general support-sixteen theorem.

The support-preserving boundary method is nevertheless sharp: at size fourteen an explicit $7+7$ state is neither directly cleanable nor circuit-deletable. It has a simple bridgeless cubic realization on 18 vertices whose displayed size-fourteen projection is uncleanable under every extension. The graph's minimum projection size is nevertheless five and all four minima are cleanable, so this is a counterexample to the unrestricted boundary dichotomy, not to the minimum-projection principle or FiveCDC. The new theorem escapes only after a support-neutral Kempe recolouring.

We also delimit what the four initial exchange inequalities can prove. A $K_4 - e$ two-pole inflation preserves the support, complement components, and all two-colour boundary path pairings, while making every non-support traversal arbitrarily long. Once every support circuit uses all four affine colours, sufficiently long inflation forces all four exchange inequalities for any displayed boundary realization. An explicit 230-vertex simple bridgeless cubic example satisfies the inequalities and defeats every one-round fixed-colour Kempe multiswitch, but reaches a strict circuit deletion in two rounds. Thus the unresolved step at this stage cannot use only the four inequalities at the initial flow. The later strict-lock theorem goes further and rules out universal repair of a globally minimum projection itself.

Two further structural results sharpen that dynamic frontier. The tensor theorem directly cleans every one-circuit support, strictly subsuming the earlier two-terminal-component argument, and gives the circuitwise-balanced several-circuit corollary above. Conversely, chains of the edge-deleted $K_{3,3}$ two-pole reflect all boundary-changing Kempe moves. They transfer any hypothetical goal-free boundary orbit to one in which all four exchange inequalities hold at every reachable state. No goal-free base orbit is known, so this conditional transfer supplies neither a counterexample nor a resolution.

We perform an exact canonical census of all 41,301 connected simple cubic graphs of order 18. Among the 39,866 bridgeless graphs, exactly 179 are not Tait-colourable, and every minimum-cardinality extendable projection on all 179 is cleanable. A second exact replay uses frozen canonically generated sources: 14,009 cyclically 4-edge-connected non-Tait records through order 28, and a separate 12,892-record hard order-22 source. All 147,539 minimum extendable projections in those files are cleanable. Completeness of the source populations is inherited from their upstream generation packages; the result on the literal frozen files is checked directly. An additional exact SAT scan covers seven retained order-34 and 7,654 retained order-40 strong-snark rows. All 433,730 minimum projections in those files have size ten and are cleanable.

The 162-vertex strict-lock graph proves that the universal statement suggested by these computations is false beyond the tested range. Its explicit FiveCDC shows exactly why this is a proof-method result: the required clean projection exists but is not globally minimum. We make no claim of a FiveCDC resolution or of priority for the flow formulation. An elementary incidence-cluster construction proves that resolving the cubic FiveCDC assertion would resolve the standard conjecture for arbitrary finite bridgeless multigraphs.

Contents

1 Status, conventions, and contribution boundary

A binary cycle of a graph is an Eulerian edge set; it may be empty or disconnected. A k-cycle double cover is an indexed family of k binary cycles in which every edge occurs exactly twice. Empty members may be appended, so "at most five" and "five, allowing empty members" are equivalent. The five-cycle double cover conjecture, attributed to Celmins and Preissmann, asserts that every bridgeless graph has such a cover with at most five members [1, 12]. The ordinary cycle double cover conjecture was proved in July 2026 by OpenAI, with independent human expositions by Oum and Geelen [11, 7, 4]. That proof produces eight labelled Eulerian subgraphs; the stronger bound of five studied here remains Conjecture 18 in Oum's exposition.

We work with the cubic flow formulation in [6, Conjecture 3.19]. All projection and boundary theorems below apply to finite loopless cubic multigraphs unless "simple" is stated. Parallel edges are distinct edge objects. A loop would contribute twice to vertex degree; loops are excluded from those theorems to keep the cut notation literal. The finite census is restricted to simple graphs.

Proposition 1.1 (standard graph-class reduction)**.** *The standard FiveCDC assertion for finite bridgeless multigraphs, with loops and parallel edges allowed, is equivalent to its restriction to finite bridgeless loopless cubic multigraphs.*

Proof. Only the reduction to the cubic class needs proof. Discard isolated vertices. At a vertex v of incidence degree $d(v)$, make a cycle Z_v of $d(v)$ new vertices and assign the incidences at v bijectively to them. When $d(v) = 2$, Z_v is the two-edge parallel cycle. Replace every original nonloop edge by an edge joining its two assigned cluster vertices. Replace an original loop by an edge joining the two distinct cluster vertices assigned its two incidences. Every cluster vertex now has two cluster-edge incidences and one old-edge incidence, so the expansion $X(G)$ is loopless and cubic.

Every cluster edge lies on Z_v. An old edge arising from a loop has an alternative path in its cluster. For a nonloop old edge $e = uv$, bridgelessness gives a u-to-v path in $G - e$; joining consecutive attachments inside the clusters lifts it to an alternative path between the ends of e in $X(G)$. Thus $X(G)$ is bridgeless.

Let $(D_1, \ldots, D_5)$ be a five-cycle double cover of $X(G)$, and let C_i consist of the old edges in D_i, identified with the original edges of G. Sum modulo two the degree equations for D_i over all vertices of Z_v. Every cluster edge is counted twice and cancels, leaving an even number of selected original incidences at v. An original loop contributes its two incidences, exactly as required. Hence every C_i is Eulerian. Exact double coverage is inherited edge by edge, so $(C_1, \ldots, C_5)$ covers G. $\square$

Remark 1.2 (reduction scope). This proposition reduces the universal graph class; it does not identify several boundary occurrences at a higher-degree vertex inside the minimum-projection argument. A separate self-contained reduction package also proves that a minimum cubic counterexample could be taken simple, non-3-edge-colourable, of girth at least five, and cyclically 4-edge-connected. These standard reductions are not used to infer the finite census populations below.

Hušek and Šámal prove that the cubic FiveCDC problem is equivalent to selecting a nowhere-zero $\mathbb{F}_2^3$-flow with an additional parity condition [6, Theorem 3.16 and Conjecture 3.19]. Their criterion is related to the prescribed-cycle formulation of Hoffmann-Ostenhof [5]. We claim no novelty or priority for either formulation. Earlier integer-flow-pair characterizations distinguish standard and orientable 5CDC conditions [14]; 4-flow and Catlin-reduction methods give further sufficient conditions and snark families [9].

The contribution of this note is narrower:

1. a minimum-support exchange theorem, with a complete proof;
2. a human proof through size five and an exact boundary extension through size fifteen, plus a sharp size-fourteen counterstate to the unrestricted support-preserving boundary method;
3. a universal tensor theorem that directly cleans every one-support-circuit state and every circuitwise-balanced several-circuit state, together with the Petersen two-occurrence boundary showing why the balance hypothesis cannot simply be dropped;
4. a two-occurrence clean-or-delete theorem covering all states through support sixteen, including interaction loops, and all loopless states through support eighteen, reduced by hand to seventeen finite interaction profiles and exactly checked;
5. an exact census of one $8 + 8$ support-sixteen word orbit and a matching-observed one-round Kempe dichotomy on its eight residuals: six universal reductions and two explicit 42-vertex Tait obstructions to that narrowly specified game;
6. an affine-XOR duality theorem for general several-circuit cleaning and a six-map identity that legally neutralizes any one chosen dual inconsistency witness, together with an exact four-step cycle proving that this move alone has no monotone termination argument;
7. a shortest-join dual theorem forcing every positive dual shore to split a complement component, with the support-density bound $|h| \leq 6|V|/7$ in cubic graphs and a complete rigidity theorem at equality;
8. a six-map colour-load inequality coupling a neutral recolouring to the dynamically recomputed exchange inequalities, with an explicit strict descent whenever the bound fails, plus a smallest cube-graph counterstate showing that local rainbow parity alone need not violate it;

9. four simultaneous shortest-join consequences;
10. a general inflation theorem showing that those four inequalities, at one fixed flow, do not force one-round Kempe descent, together with a smallest goal-free boundary survivor satisfying the load and all four static shortest-join filters;
11. an exact full-flow master formulation as a minimum over low-flow zero matchings and shortest joins, in which cleanliness is precisely a second edge-disjoint join;
12. a strict parity-lock construction of a 162-vertex graph whose unique minimum projection is uncleanable, together with an explicit five-cycle double cover of that graph;
13. a reproducible canonical order-18 census and an expanded exact replay on frozen sources through order 28.

FiveCDC itself remains open.

2 Extendable and clean Fano projections

Let $G = (V, E)$ be a loopless cubic graph and put $K = \mathbb{F}_2^2$. Write a three-bit flow as

$$f = (h, s) : E \longrightarrow \mathbb{F}_2 \times K, \tag{1}$$

where $h : E \to \mathbb{F}_2$ and $s : E \to K$ are flows. We identify a binary flow with its support. The established term *Fano flow* usually records which lines of the Fano plane occur at cubic vertices [10, 8]. Our *Fano projection* instead means one chosen coordinate support in the Hušek–Šámal criterion; no minimum-support result from the former terminology is being invoked.

Definition 2.1. A binary cycle h, possibly zero, is *extendable* if there is a flow $s : E \to K$ for which $f = (h, s)$ is nowhere zero.

Writing $s = (p, q)$, extendability has the following elementary form:

$$E - h \subseteq p \cup q, \tag{2}$$

where p and q are binary cycles. Indeed, an edge of $E - h$ has first coordinate zero and therefore needs a nonzero low coordinate.

Fix an extension $f = (h, s)$. For $c \in K$, put

$$M_c = \{e \in h : s(e) = c\}. \tag{3}$$

Proposition 2.2. *Every M_c is a matching, and the four sets M_c partition h.*

Proof. The partition statement is immediate. If two edges incident with a cubic vertex both had flow value $(1, c)$, their sum would be zero and flow conservation would force the third incident edge to have value zero, contrary to nowhere-zeroness. □

Let W be a component of the spanning subgraph $(V, E - h)$. Every edge of $\delta(W)$ belongs to h. If

$$n_c(W) = |\delta(W) \cap M_c| \pmod 2,$$

then cut conservation for the first coordinate and the two low coordinates gives

$$\sum_{c\in K} n_c(W) = 0, \qquad \sum_{c\in K} n_c(W)c = 0. \tag{4}$$

The 3×4 binary matrix in (4) has one-dimensional kernel, generated by the all-ones vector. Hence the four parities $n_c(W)$ are equal.

Definition 2.3. An extension is *clean* if $n_c(W) = 0$ for every component W of $(V, E-h)$, for one, and hence every, $c \in K$. A projection h is *cleanable* if it has a clean extension. A component with all four parities equal to one is *rainbow-odd.*

If $s = (p, q)$, the class M_{11} on $\delta(W)$ is $\delta(W) \cap p \cap q$. Thus cleanability is exactly the existence of binary cycles p, q satisfying (2) and

$$|\delta(W) \cap p \cap q| \equiv 0 \pmod 2 \tag{5}$$

for every component W of $(V, E-h)$. By [6, Theorem 3.16], existence of such an h is equivalent to the cubic FiveCDC assertion. In the Tait-colourable case one may take $h = \varnothing$ and a nowhere-zero K-flow.

3 The minimum-projection exchange theorem

Assume now that G is not Tait-colourable, so an extendable projection cannot be empty. Choose h of minimum cardinality among all extendable projections, and fix any extension $f = (h, s)$.

Theorem 3.1 (minimum-projection exchange)**.** *For every $c \in K$ and every binary cycle C with $C \cap M_c = \varnothing$,*

$$|C \cap h| \le |C - h|. \tag{6}$$

Equivalently, h is a minimum-cardinality binary cycle containing M_c, simultaneously for all four $c \in K$.

Proof. Fix $c \in K$, put $t = (1, c)$, and add t to the flow value on every edge of C:

$$f'(e) = \begin{cases} f(e) + t, & e \in C, \\ f(e), & e \notin C. \end{cases} \tag{7}$$

Since C is a binary cycle, the added function is a flow. The only edges that could become zero in (7) are those whose old value was t, and these are exactly the edges of M_c. The hypothesis $C \cap M_c = \varnothing$ therefore makes f' nowhere zero.

The first coordinate of f' is $h \triangle C$. This is another extendable projection, so minimality gives

$$|h| \le |h \triangle C| = |h| + |C - h| - |C \cap h|.$$

This is (6).

For the equivalent statement, rebase the low coordinates by replacing s with the flow $s + ch$. Its exact zero set is M_c: outside h, the low value was already nonzero, while on h it vanishes exactly when $s(e) = c$.

Let h' be any binary cycle containing M_c. The flow $(h', s + ch)$ is nowhere zero. On an edge outside h', its low value is nonzero because every zero lies in $M_c \subseteq h'$; on an edge of h', its first coordinate is one. Thus h' is extendable, and $|h| \le |h'|$.

Conversely, if $C \cap M_c = \varnothing$, then $h \triangle C$ contains M_c. Applying the minimum-containing property to $h \triangle C$ recovers (6). ∎

3.1 Shortest joins and four-colour consequences

Let T_c be the set of endpoints of M_c. Removing M_c from a binary cycle containing M_c produces a T_c-join in $G - M_c$, and adjoining M_c reverses the correspondence. Therefore Theorem 3.1 has the following form.

Corollary 3.2. *For every $c \in K$, the edge set*

$$J_c = h - M_c$$

is a shortest T_c-join in $G - M_c$.

This is not the already-refuted principle that a cardinality-minimum exact-zero matching must extend to a FiveCDC certificate. Here the objective is the cardinality of the whole binary cycle $M_c \cup J_c$, and the same projection is optimal after all four affine rebases.

Proposition 3.3 (zero dual price on quotient shores)**.** *Fix $c \in K$. In $G - M_c$, the standard odd-cut dual for the minimum T_c-join problem is*

$$\begin{aligned} &\textit{maximize} && \sum_{\substack{S \subseteq V \\ |S \cap T_c| \text{ odd}}} y_{c,S} \\ &\textit{subject to} && \sum_{\substack{S:\ e \in \delta(S) \\ |S \cap T_c| \text{ odd}}} y_{c,S} \le 1 \quad (e \in E - M_c), \qquad y_{c,S} \ge 0. \end{aligned} \tag{8a}$$

Its optimum is $|J_c| = |h| - |M_c|$. In every optimal dual,

$$y_{c,S} = 0 \tag{8b}$$

whenever S is a union of components of $G - h$. Consequently every positive dual shore splits the interior of at least one component of $G - h$.

Proof. The Edmonds–Johnson T-join polyhedron [2] and Corollary 3.2 give the displayed optimum with χ_{J_c} as an optimal primal solution. Complementary slackness therefore says

$$y_{c,S} > 0 \quad \Longrightarrow \quad |J_c \cap \delta(S)| = 1. \tag{8c}$$

Now suppose S is a union of components of $G - h$. Its cut lies inside h. Put

$$n_d = |\delta(S) \cap M_d| \pmod 2 \qquad (d \in K).$$

Cut conservation for the three flow coordinates says $\sum_d n_d = 0$ and $\sum_d n_d d = 0$. The resulting 3×4 binary matrix has kernel generated by the all-ones vector, so the four bits n_d are equal.

If they are all zero, then $|S \cap T_c| \equiv |\delta(S) \cap M_c| \equiv 0$, and S does not index a variable in (8a). If they are all one, then S is T_c-odd but

$$|J_c \cap \delta(S)| = \sum_{d \neq c} |\delta(S) \cap M_d| \ge 3, \tag{8d}$$

contradicting (8c) if $y_{c,S} > 0$. Thus (8b) holds in every optimal dual. A shore that splits no component of $G - h$ is precisely such a component union. $\square$

Corollary 3.4 (support-density bound)**.** *For every $c \in K$,*

$$|E - h| \geq |J_c| = |h| - |M_c|. \tag{8e}$$

Hence

$$|E - h| \geq \tfrac{3}{4}|h|, \qquad \textit{and in a cubic graph} \qquad |h| \leq \tfrac{6}{7}|V|. \tag{8f}$$

Proof. Fix an optimal dual in (8a). Every positive dual shore contains exactly one J_c-edge by (8c), and at least one complement edge by Proposition 3.3; thus its cut in $G - M_c$ has size at least two. If $\ell_c(e) \leq 1$ is the load on edge e, then

$$2|J_c| \leq \sum_S y_{c,S}|\delta_{G-M_c}(S)| = \sum_{e \in E - M_c} \ell_c(e) \leq |J_c| + |E - h|. \tag{8g}$$

The last inequality uses complementary-slackness tightness on every edge of J_c and unit capacity on every complement edge. This proves (8e). Choose c with $|M_c| \leq |h|/4$ to get the first inequality in (8f). Finally substitute $|E| = 3|V|/2$. □

Proposition 3.5 (rigidity at density equality)**.** *Put $m = |E - h|$, fix any optimal duals in (8a), and define*

$$\begin{aligned} A_c &= \sum_S y_{c,S}\big(|\delta_{G-M_c}(S)| - 2\big), \\ U_c &= \sum_{e \in E - h} \big(1 - \ell_c(e)\big). \end{aligned} \tag{8h}$$

Then

$$4m - 3|h| = \sum_{c \in K} (A_c + U_c), \tag{8i}$$

and every term on the right is nonnegative.

If equality holds in (8f), equivalently $4m = 3|h|$, then:

1. *every positive dual shore has exactly one J_c-edge and one complement edge in $G - M_c$;*
2. *every complement edge is saturated in all four duals, and $G - h$ is a forest;*
3. *$|h|$ is divisible by twelve,*

$$|V| - |h| = \frac{|h|}{6}, \qquad \#\,\mathrm{comp}(G - h) = \frac{5|h|}{12}; \tag{8j}$$

4. *each nonzero low colour occurs on exactly $|h|/4$ complement edges and at exactly $|h|/3$ support-boundary incidences; and*
5. *every six-map-integrable component union, in particular every affine-cleaning witness shore, has equal selected derivative counts in the three nonzero colours, and equal outside counts.*

Consequently an uncleanable globally minimum projection at density equality must have $|h| \geq 24$.

Proof. By (8c), every positive shore has exactly one J_c-edge and, by Proposition 3.3, at least one complement edge. Thus $A_c \geq 0$; unit capacity gives $U_c \geq 0$. Double-counting weighted cut-edge incidences gives

$$2|J_c| + A_c = \sum_S y_{c,S}|\delta_{G-M_c}(S)| = \sum_{e \in E-M_c} \ell_c(e) = |J_c| + m - U_c. \tag{8k}$$

Hence $m - |J_c| = A_c + U_c$. Summing over the four colours and using $\sum_c |J_c| = 3|h|$ proves (8i).

At equality every A_c and U_c vanishes. The first two assertions about dual cuts and saturation follow immediately. For any complement edge e, saturation supplies a positive shore whose only complement crossing is e. Such a shore cannot cut a complement circuit in just one edge, so every complement edge is a bridge of $G - h$, proving the forest assertion.

Let m_t count complement edges of nonzero low colour t. The functional half-load argument in the proof of Proposition 3.24 gives

$$m - m_t \geq \frac{|h|}{2}.$$

Since $m = 3|h|/4$ and $\sum_t m_t = m$, all three m_t equal $|h|/4$. Put $n = |V| - |h|$. Cubicity gives $n = |h|/6$. If k_t is the number of colour-t complement incidences at support vertices, then

$$2m_t = n + k_t,$$

because every off-support cubic vertex sees the three nonzero colours once each. Hence $k_t = |h|/3$, proving divisibility by twelve.

If the complement forest has q components, its support vertices are its leaves and its off-support vertices have degree three. Summing the tree identity "leaves = internal vertices +2" gives $n = |h| - 2q$, hence $q = 5|h|/12$.

Finally let a_t be the three selected derivative counts of a six-map-integrable component union. The outside counts are $|h|/3 - a_t$, and (15m) gives

$$\max_t a_t + \max_t\bigl(|h|/3 - a_t\bigr) \leq 2n = |h|/3.$$

Thus $\max_t a_t = \min_t a_t$, proving exact balance inside and outside. The only positive multiple of twelve below twenty-four is twelve, and Theorem 3.50 excludes an uncleanable minimum projection of that size. $\square$

Remark 3.6. The full proof, two arithmetic audits, frozen outputs, and hash ledger are in the density-equality-rigidity package under `scratch/`. Strict density slack $4m - 3|h| > 0$ remains open.

Corollary 3.7 (flow-resistance bound)**.** *Let $r_f(G)$ be the minimum number of zero edges in an $\mathbb{F}_2^2$-flow on G. Every extendable projection satisfies*

$$|h| \geq 4r_f(G). \tag{8}$$

Proof. For each c, the rebased low flow $s + ch$ has exact zero set M_c. Hence $|M_c| \geq r_f(G)$. The four M_c's partition h. $\square$

Corollary 3.8 (four-colour circuits)**.** *Every circuit component of h contains at least one edge of each of the four sets M_c.*

Proof. If a circuit component D omitted M_c, then D would be a binary cycle avoiding M_c, but $|D \cap h| = |D| > |D - h| = 0$, contrary to (6). $\square$

Theorem 3.9 (minimum projections through size five)**.** *Let G be a connected bridgeless loopless cubic graph; parallel edges are allowed. If G is not Tait-colourable and its minimum extendable projection has size at most five, then every minimum extendable projection is cleanable. If G is Tait-colourable, the zero projection is a clean minimum.*

Proof. The Tait case follows by taking a nowhere-zero K-flow as the low coordinate: its zero affine class is empty, so the equal cut parities in (4) are all even. Assume henceforth that G is non-Tait, choose a minimum extendable h, and fix an arbitrary extension $f = (h, s)$.

By Corollary 3.8, every circuit component of h meets all four M_c. Since G is loopless, $|h| \leq 5$ therefore forces h to be one ordinary circuit D of length $n = 4$ or 5; a parallel two-circuit cannot meet four classes. Write

$$v_0, e_0, v_1, e_1, \ldots, v_{n-1}, e_{n-1}, v_0, \qquad e_i = v_i v_{i+1},$$

cyclically, and put $c_i = s(e_i)$. Adjacent c_i's are distinct: otherwise the third edge at their common vertex would have full flow value zero.

Let $K_0 = G - h$. Every component of K_0 contains a vertex of D, by connectedness. Suppose the extension is dirty. For a dirty component W, membership along D toggles oddly on each of the four affine colour classes.

If $n = 4$, the four colours occur once each, so membership toggles on every e_i. The components of K_0 are exactly the two alternating vertex pairs. Indeed, separating either pair further would create a component whose cut sees exactly two affine colours oddly, contrary to (4).

If $n = 5$, rotate and rename the colours so their cyclic word is A, B, A, C, D. Put

$$z_i = 1[v_i \in W], \qquad y_i = z_i + z_{i+1}.$$

Dirtiness says

$$y_1 = y_3 = y_4 = 1, \qquad y_0 + y_2 = 1.$$

Up to complement, the solutions are $W = \{v_2, v_4\}$ and $W = \{v_1, v_4\}$. Summing the common odd cut parity over all K_0-components shows that the number of dirty components is even. The only component-compatible shore disjoint from either displayed dirty shore is its complement: exhaustive substitution in the preceding four equations also gives the clean pair $\{v_1, v_2\}$ and its complement, both of which intersect either dirty partition. Hence K_0 again has exactly two components.

Let k_i be the unique K_0-edge incident with v_i, and put

$$d_i = s(k_i) = c_{i-1} + c_i \neq 0.$$

On each of the two K_0-components, apply an independently chosen element of $\mathrm{GL}(2, 2)$ to every low flow value. This preserves internal conservation and nonzeroness. We now choose nonzero values r_i on e_i so that

$$r_{i-1} + r_i = g_i, \tag{9}$$

where g_i is the transformed value on k_i.

For $n = 4$, the boundary values on each alternating pair agree: $d_0 = d_2 = x$ and $d_1 = d_3 = y$. Map x and y, independently, to the same nonzero α. If $\{\alpha, \beta, \gamma\} = K - \{0\}$, assigning β, γ alternately to the edges of D satisfies (9).

For $n = 5$, translate the affine colour names so $A = 0$, write $B = x, C = y, D = x + y$, and normalize $x = 1, y = 2, x + y = 3$. Thus

$$d = (3, 1, 1, 2, 1).$$

The complete repairs for the two component partitions are

K_0-components on v_i	$(g_0,\dots,g_4)$	$(r_0,\dots,r_4)$
$\{0,1,3\} \mid \{2,4\}$	$(3,1,2,2,2)$	$(2,3,1,3,1)$
$\{0,2,3\} \mid \{1,4\}$	$(3,3,1,2,3)$	$(1,2,3,1,2)$.

In each row the three-terminal component uses the identity. On the two-terminal component, both original boundary values are 1, which an invertible linear map sends respectively to 2 or 3. Direct cyclic substitution verifies (9).

We have constructed a nowhere-zero K-flow on every edge of G. At a cubic vertex its three incident values are therefore the three distinct nonzero elements of K, giving a Tait colouring. This contradicts the hypothesis. Thus the arbitrary extension of h was clean, and in particular h is cleanable. □

Remark 3.10. The finite case split and the two repair rows are independently replayed by `verify.py` in the bundled size-five theorem package.

3.2 A line-switch lemma and the size-six/seven boundary

We need one support-preserving fact. Identify the seven nonzero full flow values with $1,\dots,7$ under bitwise addition, so that

$$L = \{1,2,3\} = \{(0,t) : t \in K - \{0\}\}, \qquad A = \{4,5,6,7\} = \{(1,c) : c \in K\}.$$

Let $\mathcal{K}$ be the components of $G-h$, and let $r \in \mathbb{F}_2^{\mathcal{K}}$ indicate the rainbow-odd components. For $t \in L$, put $N_t = f^{-1}(t)$ and $P_t = \{4, 4+t\}$. If C is a binary cycle of $G - N_t$, define

$$\tau_t(C)_W = |C \cap \delta(W) \cap f^{-1}(P_t)| \pmod 2 \qquad (W \in \mathcal{K}). \tag{10}$$

Adding t along C is a valid line-preserving switch: avoiding N_t prevents a zero value, and (10) is its change to the defect vector.

Proposition 3.11 (combined-line span)**.**

$$r \in \sum_{t\in L} \operatorname{im}\tau_t. \tag{11}$$

Consequently, if $G - h$ has at most two components, then h is cleanable by at most one valid line-preserving switch.

Proof. Suppose (11) fails. There is a vector $y \in \mathbb{F}_2^{\mathcal{K}}$ with

$$y \cdot r = 1, \qquad y \cdot \tau_t(C) = 0 \quad (t \in L,\ C \text{ a cycle of } G - N_t).$$

Let U be the union of the components selected by y, and let $u : V(G) \to \mathbb{F}_2$ indicate U. Cycle–cut orthogonality says that

$$R_t = \delta(U) \cap f^{-1}(P_t)$$

is a cut in $G - N_t$. Choose a vertex indicator x_t for that cut, and write $q = (u, x_1, x_2, x_3)$.

Define

$$\mathcal{A}(q) = \big(u(x_2+x_3),\ u(x_1+x_3),\ x_1x_2 + x_1x_3 + x_2x_3\big) \in \mathbb{F}_2^3.$$

Across an edge, the only possible nonzero differences of $(u; x_1x_2x_3)$, indexed by its flow value, are

$f(e)$	1	2	3	4	5	6	7
Δq	$(0;100)$	$(0;010)$	$(0;001)$	$(1;111)$	$(1;100)$	$(1;010)$	$(1;001)$.

The zero difference is also allowed in every column. Direct substitution gives the pointwise identity

$$\big(\mathcal{A}(q_v) + \mathcal{A}(q_w)\big) \cdot f(vw) = \begin{cases} 1, & f(vw) = 4 \text{ and } u(v) \neq u(w), \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

Summing (12) over all edges and regrouping at vertices makes its left side

$$\sum_v \mathcal{A}(q_v) \cdot \left(\sum_{e \ni v} f(e) \right) = 0$$

by flow conservation. The right side is $|\delta(U) \cap f^{-1}(4)| = y \cdot r = 1$, a contradiction. This proves (11).

Every vector in every image of τ_t has even Hamming weight, since an affine edge between two $G - h$ components is counted at both ends. For one component the even-weight space is zero. For two components it is one-dimensional; if $r \neq 0$, (11) forces one of the three image spaces to contain its unique nonzero vector. The corresponding single valid switch kills r. A single switch has no mixed-switch quadratic correction, completing the proof. □

Theorem 3.12 (minimum projections through size seven)**.** *Under the hypotheses of Theorem 3.9, every minimum extendable projection of size at most seven is cleanable.*

Proof. Only sizes six and seven remain. The four-colour circuit corollary forces h to be one circuit

$$v_0, e_0, v_1, e_1, \ldots, v_{n-1}, e_{n-1}, v_0, \qquad n \in \{6, 7\}.$$

Put $c_i = s(e_i)$, and encode the component of $G - h$ containing v_i by π_i. The c_i's form a proper cyclic word using all four affine colours. Conservation on every component W gives

$$\sum_{\pi_i = W} (c_{i-1} + c_i) = 0,$$

and its four affine cut parities are either all zero or all one.

The exact finite boundary classifier enumerates every such word and every set partition π. For each state it also enumerates one independent element of $\mathrm{GL}(2,2)$ on the low flow values of every component and solves the n cycle equations. Its complete counts are

n	valid dirty pairs	failed linear repairs	failures with more than two components
6	2712	432	0
7	26880	5376	0

Every dirty state either produces a nowhere-zero K-flow on all of G, contradicting the non-Tait hypothesis, or has exactly two components of $G-h$. At size six the failed states may additionally be quotiented by the affine group on the four colours, the dihedral group of the cycle, and component relabelling; they give the three representatives

$(c_0, \ldots, c_5)$	$(\pi_0, \ldots, \pi_5)$
010123	001001
010232	001010
012013	000101

There are 144 labeled states per orbit. At size seven there is no residual normal form after the two-component theorem. The dependency-free enumerator regenerates all words, partitions, conservation tests, dirty states, and linear repairs rather than trusting the stored counts.

Proposition 3.11 makes every failed state clean by one valid switch. Hence no dirty size-six or size-seven global minimum is uncleanable. □

Remark 3.13. The exact classifications, proof of semantics, size-six representatives, and full standalone span proof are bundled in the cumulative through-seven package. This is a finite boundary theorem, not a search over sampled graphs.

3.3 Direct component repair through size ten

The preceding Tait repair required every new low value to be nonzero. For the original three-bit flow this is unnecessarily restrictive on h: an edge of h has first coordinate one and therefore remains nonzero even when its low value is zero. Exploiting this observation gives a stronger repair.

Proposition 3.14 (direct boundary repair certificate). *Let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2 \times K$-flow. Index the vertices and edges of each circuit of h cyclically, put*

$$c_i = s(e_i), \qquad d_i = c_{i-1} + c_i,$$

and let π_i denote the component W_{π_i} of $G - h$ containing the circuit vertex v_i. Suppose there are maps $L_a \in \mathrm{GL}(2, 2)$ and values $r_i \in K$ such that

$$r_{i-1} + r_i = L_{\pi_i} d_i \tag{13}$$

on every circuit, and, for every component W_a and every $c \in K$,

$$\bigl|\{e_i \in \delta(W_a) : r_i = c\}\bigr| \equiv 0 \pmod 2. \tag{14}$$

Then h is cleanable.

Proof. Apply L_a to the low value of every edge in W_a. At a vertex outside h, all three incident edges lie in the same component, so linearity preserves the flow equation. At a circuit vertex v_i, the unique edge outside h receives value $L_{\pi_i} d_i$, and (13) is exactly the missing vertex equation. Every edge outside h retains a nonzero low value because L_a is invertible. Every edge of h has full value $(1, r_i)$, which is nonzero even if $r_i = 0$. The repaired flow is therefore nowhere zero. Equation (14) is precisely the clean affine-class cut condition, component by component. □

Theorem 3.15 (minimum projections through size ten). *Let G be a connected bridgeless loopless cubic graph; parallel edges are allowed. Every cardinality-minimum extendable projection of size at most ten is cleanable.*

Proof. The zero projection is clean. For a nonzero minimum projection, Corollary 3.8 says that every circuit component uses all four affine colours and hence has length at least four. The possible length partitions through total size ten are therefore

$$\begin{gathered} 4, 5, 6, 7, 8, 9, 10, \\ 4+4, \quad 4+5, \quad 4+6, \quad 5+5. \end{gathered} \tag{15}$$

For a fixed shape, the exact checker enumerates every proper cyclic word on K using all four colours on every circuit, modulo the global affine action and the exact dihedral action on each circuit (including circuit interchange when the lengths agree). For every canonical word it enumerates every set partition π of the circuit vertices. It retains exactly those partitions satisfying component conservation

$$\bigoplus_{\pi_i=a} d_i = 0$$

and the four equal affine cut parities, with at least one dirty component. It then exhausts all six choices of L_a independently for every component and all four starting values independently on every circuit; equation (13) forces the remaining r_i's. Finally it tests (14) literally.

The complete counts are

shape	canonical words	valid dirty states	failed direct repairs
4	1	1	0
5	1	2	0
6	5	27	0
7	7	116	0
8	26	1386	0
9	49	8840	0
10	151	102,290	0
$4+4$	2	114	0
$4+5$	2	372	0
$4+6$	11	7798	0
$5+5$	6	4232	0

These are all shapes in (15), and every retained boundary state has a certificate of Proposition 3.14. Hence every minimum projection in the stated range is cleanable. □

Remark 3.16. The dependency-free replay is the `verify.py` program in the bundled through-ten direct-cleaning package. It regenerates the words, partitions, conservation tests, and repairs. A separately written through-nine package agrees on the overlapping range, and two independent agent runs reproduced the size-ten counts. These checks are not independent human peer review.

3.4 The first direct-repair failures and size eleven

The direct-cleaning statement in Theorem 3.15 is not universal. A second elementary certificate explains why its first failures nevertheless cannot be globally minimum.

Proposition 3.17 (circuit-deletion certificate). *Let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2 \times K$-flow, and let D be a circuit component of h. Suppose component maps L_a and values r_i satisfy the vertex equations (13), and suppose $r_i \neq 0$ on every edge of D. Then $h - D$ is extendable.*

Proof. Use first coordinate one on $h - D$ and zero on D. On all circuit edges use the low values r_i, and inside each component of $G - h$ use $L_a s$. The proof of Proposition 3.14 verifies all vertex equations and nonzeroness off D. On D, the first coordinate is now zero but the assumed low value is nonzero. The result is therefore a nowhere-zero flow with first-coordinate support $h - D$. □

Proposition 3.18 (fixed-map affine cleaning). *Fix component maps L_a for which the circuit equations (13) are consistent, and fix one solution r^0. Let $\mathcal{D}$ be the set of circuit components of h. Every solution is of the form*

$$r_i = r_i^0 + z_D \qquad (e_i \in D),$$

with one independently chosen translation $z_D \in K$ per circuit. For a component W_a, define

$$b_a = \sum_{e_i \in \delta(W_a)} q(r_i^0), \qquad t_{aD} = \bigoplus_{e_i \in \delta(W_a) \cap D} r_i^0,$$

where $q(x_1, x_2) = x_1 x_2$, and put $B(x, y) = x_1 y_2 + x_2 y_1$. Then the repaired extension is clean exactly when

$$b_a + \sum_{D \in \mathcal{D}} B(t_{aD}, z_D) = 0 \qquad \text{for every } a. \tag{15a}$$

In particular, cleanliness for fixed component maps is an affine linear system over $\mathbb{F}_2$, with at most $2(|\mathcal{D}| - 1)$ scalar variables.

Proof. The difference of two solutions of (13) is equal on consecutive edges, and hence is constant around each circuit. Every circuit meets $\delta(W_a)$ evenly. Summing (13) at all circuit vertices in W_a also gives $\bigoplus_{e_i \in \delta(W_a)} r_i = 0$. Consequently the four parities in (14) are either all zero or all one: their total is zero, and their two K-coordinate sums are zero.

The common defect bit equals $\sum_{e_i \in \delta(W_a)} q(r_i)$. Indeed, $\mathbf{1}_{x=0} = 1 + x_1 + x_2 + q(x)$, and the constant and linear sums vanish by the preceding two parity identities. Substitute $r_i = r_i^0 + z_D$ and use

$$q(x + z) = q(x) + q(z) + B(x, z).$$

The $q(z_D)$ term occurs an even number of times on each circuit cut and cancels, leaving exactly (15a). Finally $\bigoplus_D t_{aD} = 0$, so translating every circuit by the same element changes no equation; one z_D may be fixed to zero. □

Remark 3.19. This proposition is a human-checkable reduction of the translation search, not a universal solution. If its affine system is inconsistent, binary linear duality gives a short XOR witness y with $y^\mathsf{T}\Phi = 0$ and $y^\mathsf{T} b = 1$. The exact remaining issue is to choose the component maps so that either this system is consistent or one integrated circuit word omits a colour, which triggers Proposition 3.17.

Proposition 3.20 (dual-cut feasible move)**.** *If the fixed-map system (15a) is inconsistent, there is a nonempty proper union Y of components of $G - h$ such that, for every circuit D of h,*

$$\bigoplus_{e_i \in \delta(Y) \cap D} r_i^0 = 0, \tag{15b}$$

while an odd number of the circuits meet $\delta(Y)$ oddly in each of the four low colours. Moreover, for every $T \in \mathrm{GL}(2, 2)$, postcomposing all L_a with T for $W_a \subseteq Y$, and changing no other component map, preserves circuit consistency.

Proof. Let y be a dual XOR witness from the preceding remark and let Y be its selected component union. Summing t_{aD} over the selected rows cancels support edges internal to Y twice and leaves exactly the left side of (15b). Nondegeneracy of B and $y^\mathsf{T}\Phi = 0$ make this xor zero for every circuit. Each circuit crosses $\delta(Y)$ evenly, so zero colour xor forces its four cut-colour parities to be equal. The identity $y^\mathsf{T} b = 1$ says that the sum of their q-parities is odd; hence an odd number of circuits are rainbow-odd on the cut.

Finally, summing (13) over the circuit vertices in Y gives zero by (15b). Postcomposition sends this partial sum to $T(0) = 0$; the complementary partial sum is also zero because the original circuit total was zero. Thus every circuit equation remains consistent. □

Remark 3.21. The proposition turns every failed translation system into six coordinated feasible map choices. The next proposition proves that these choices can always neutralize the particular dual witness that produced them. A universal proof still needs a well-founded progress statement, because a different witness may appear after the switch.

Proposition 3.22 (six-map witness neutralization)**.** *Let Y be the component union supplied by Proposition 3.20. For $U \in \mathrm{GL}(2,2)$, postcompose by U the component maps inside Y, reintegrate on every support circuit, and let*

$$F_Y(U) = \sum_{W_a \subseteq Y} b_a(U) = \sum_{e \in \delta(Y) \cap h} q(r_e^U). \tag{15c}$$

The value is independent of the permitted circuit translations. If $\{I, \rho, \rho^2\}$ is the order-three subgroup of $\mathrm{GL}(2,2)$ and τ_1, τ_2, τ_3 are its three involutions, then

$$\begin{gathered} F_Y(I) + F_Y(\rho) + F_Y(\rho^2) = 0, \\ F_Y(\tau_1) + F_Y(\tau_2) + F_Y(\tau_3) = 0. \end{gathered} \tag{15d}$$

Consequently, if Y is a dual inconsistency witness before the switch, so $F_Y(I) = 1$, then one of the six legal choices of U preserves the dual balance of Y but changes its right-hand-side sum to zero.

Proof. It is enough to work on one oriented support circuit. At its i-th vertex put $\epsilon_i = 1$ when the containing complement component lies in Y, and put

$$g_i = L_{\pi_i} d_i = r_{i-1} + r_i.$$

The dual balance in Proposition 3.20 is equivalently

$$\bigoplus_{\epsilon_i = 1} g_i = 0. \tag{15e}$$

Choose cyclic edge potentials $p_i \in K$ satisfying

$$p_{i-1} + p_i = \epsilon_i g_i. \tag{15f}$$

Equation (15e) is exactly the condition that this integration closes. After the maps inside Y are postcomposed by U, one compatible support word is

$$r_i^U = r_i + (U + I) p_i. \tag{15g}$$

Indeed its derivative is $U g_i$ at a selected vertex and g_i otherwise. The selected derivative xor after the switch is $U \bigoplus_{\epsilon_i=1} g_i = 0$; thus the balance of Y persists. Proposition 3.18 then also shows that $F_Y(U)$ is unchanged by arbitrary independent translations of the support circuits.

It remains to sum $q(r_i^U)$ on the circuit edges crossing between selected and unselected vertices. Identify K with $\{0,1,2,3\}$, with $q(3) = 1$ and $q(0) = q(1) = q(2) = 0$. Direct evaluation of the four possible r, p gives, for either of the three-element sets of maps in (15d),

$H(r,p)$	$p=0$	$p=1$	$p=2$	$p=3$
$r=0$	0	1	1	0
$r=1$	0	1	0	1
$r=2$	0	0	1	1
$r=3$	1	1	1	1

$$H(r,p) = q(r+p) + \mathbf{1}_{p \neq 0}, \tag{15h}$$

where

$$H(r,p)=\sum_{U\in\{I,\rho,\rho^2\}} q(r+(U+I)p)=\sum_{U\in\{\tau_1,\tau_2,\tau_3\}} q(r+(U+I)p).$$

Hence either three-map sum of the present circuit's contribution to F_Y is

$$\sum_{e_i\in\delta(Y)} \big(q(r_i+p_i)+\mathbf{1}_{p_i\neq 0}\big). \tag{15i}$$

Put $b_i=r_i+p_i$. Equations (15f) and $r_{i-1}+r_i=g_i$ show that b_i is constant across each consecutive run of selected vertices. Its two $q(b_i)$ terms on the flanking cut edges therefore cancel. Likewise p_i is constant across every run of unselected vertices, so the two flanking $\mathbf{1}_{p_i\neq 0}$ terms cancel. Thus (15i) is zero on every circuit, proving (15d).

Finally $F_Y(I)=1$ and the first identity in (15d) force at least one of $F_Y(\rho),F_Y(\rho^2)$ to be zero. The post-switch balance already proved means that the same selected row combination remains in the left kernel, now with zero right-hand side. It is therefore no longer an inconsistency witness. □

Corollary 3.23 (dual invisibility of neutralization)**.** *Assume h is globally minimum. For the witness shore Y in Proposition 3.22, all four old shortest-join duals have*

$$y_{c,Y}=0 \qquad (c\in K). \tag{15j}$$

After a neutralizing switch, Y is T_c-even for every recomputed colour class and disappears from all four dual programs. Moreover, for every legal map choice U,

$$\sum_{c\in K}\max D_c(U)=3|h|. \tag{15k}$$

Thus neither the selected cut coefficient nor the aggregate four-dual optimum can be a decreasing neutralization potential.

Proof. The shore Y is a union of components of $G-h$, so (15j) is Proposition 3.3. Before the switch its four cut parities are all one; after neutralization they are all zero, giving the second assertion. The switch keeps the projection h and produces another extension. Its four classes M_c^U partition h, while the corresponding optimal join has size $|h|-|M_c^U|$. Summing over c gives $4|h|-|h|=3|h|$. □

Proposition 3.24 (six-map colour-load inequality)**.** *Assume h is globally minimum. Let Y be a union of components of $G-h$ such that postcomposing the low flow on Y by every $U\in \mathrm{GL}(2,2)$ is integrable on the support circuits. For nonzero $a,b\in K$, let N_b be the number of colour-b complement edges inside Y, let O_a be the number of colour-a complement edges outside Y, and put $m=|E-h|$. Then*

$$O_a+N_b\le m-\frac{|h|}{2} \qquad (a,b\neq 0). \tag{15l}$$

If (15l) fails, its proof explicitly constructs a legal common map switch and a binary cycle giving a strict projection descent.

Equivalently, let $k_b(Y)$ count support vertices in Y whose unique incident complement edge has low colour b. Then

$$k_a(\overline{Y})+k_b(Y)\le 2\big(|V|-|h|\big) \qquad (a,b\neq 0). \tag{15m}$$

Proof. For a nonzero functional $\alpha : K \to \mathbb{F}_2$, both

$$C_\alpha = \{e : \alpha(s_e) = 1\}, \qquad C'_\alpha = h \triangle C_\alpha$$

are binary cycles. Put

$$a_\alpha = |C_\alpha \cap h|, \qquad m_\alpha = |C_\alpha - h|.$$

The first cycle avoids each M_c with $\alpha(c) = 0$, while the second avoids each M_c with $\alpha(c) = 1$. Applying the minimum-projection exchange inequality to both cycles gives

$$a_\alpha \leq m_\alpha, \qquad |h| - a_\alpha \leq m_\alpha,$$

and hence $m_\alpha \geq |h|/2$.

Fix $a, b \neq 0$, choose the functional with $\ker \alpha = \{0, a\}$, and choose $U \in \mathrm{GL}(2,2)$ with $U(b) = a$. After switching Y, the number of complement edges on which α is one is exactly

$$m_\alpha^U = m - O_a - N_b.$$

The switched flow has the same globally minimum projection, so the preceding half-load inequality applies again and proves (15l). If (15l) fails, then $m_\alpha^U < |h|/2$. According as $a_\alpha^U > |h|/2$ or not, C_α^U or $h \triangle C_\alpha^U$ uses more support than complement edges and avoids two entire low-colour classes. Either avoided class therefore certifies a legal strict descent.

For the boundary form, let n_Y be the number of off-support vertices inside Y. Every such cubic vertex sees the three distinct nonzero low colours, while every support vertex has one complement incidence. Double-counting colour incidences gives

$$2N_b = n_Y + k_b(Y), \qquad 2O_a = n_{\overline{Y}} + k_a(\overline{Y}).$$

Since h is 2-regular on its support,

$$n_Y + n_{\overline{Y}} = |V| - |h|, \qquad 2m = 3(|V| - |h|) + |h|.$$

Substitution in (15l) is exactly (15m). □

Proposition 3.25 (global minimality is essential for load forcing)**.** *There is a simple connected bridgeless planar cubic graph carrying a nowhere-zero $\mathbb{F}_2^3$-flow, a 6-circuit projection h, and a component-union shore Y with all of the following properties:*

1. *the four low-colour classes cross $\delta(Y)$ oddly;*
2. *all six common $\mathrm{GL}(2,2)$ switches on Y are integrable;*
3. *every support colour occurs; and*
4. *all nine boundary-load inequalities (15m) hold.*

The graph is Tait-colourable, so the displayed h is not globally minimum. Thus rainbow parity, cyclic closure, and minimally connected complement components do not imply a violation of (15m) without the global-minimum hypothesis.

Proof. Take vertices $0, \dots, 7$, support circuit

$$01, 12, 23, 34, 45, 50,$$

and complement edges

$$06, 26, 46, \qquad 17, 37, 57.$$

This is the cube graph, with canonical graph6 string `Gs@ipo`. In the displayed edge order, take first coordinate

$$111111000000$$

and low coordinate

$$102030123123.$$

Direct xor at every vertex is zero and no full edge value vanishes. Removing h leaves the two 3-stars

$$W_0 = \{0, 2, 4, 6\}, \qquad W_1 = \{1, 3, 5, 7\}.$$

For $Y = W_1$, the derivative word around the support circuit is `112233`, while the shore word is `010101`. Each shore half has derivative xor $1+2+3 = 0$, so every common linear switch is integrable.

The cut is the whole support circuit. Its support-colour counts are $(3, 1, 1, 1)$, hence all four are odd. Inside and outside Y, the three nonzero complement-colour counts are both $(1, 1, 1)$. Thus

$$O_a + N_b = 2 \leq |E - h| - \frac{|h|}{2} = 3$$

for every $a, b \neq 0$; equivalently, $k_a(\overline{Y}) + k_b(Y) = 2 \leq 4 = 2(|V| - |h|)$. Finally the edge-colour word

$$121323231321$$

is a Tait colouring in the same edge order. Its associated nowhere-zero K-flow has first projection zero, proving that the displayed size-six projection is not minimum. □

Remark 3.26. In the one-circuit model with two minimally connected cubic-tree components, lengths four and five force a load violation, whereas the displayed length-six state is safe. An exact boundary census gives $12, 120, 960$ states at lengths $4, 5, 6$, with respectively $0, 0, 888$ load-safe rows. The rainbow-load counterstate package under `scratch/` contains the literal graph, full six-map table, independent checker, outputs, and hash ledger.

Remark 3.27. Proposition 3.22 removes any *chosen* Farkas witness at every boundary size, but it does not remove all witnesses simultaneously. The size-fourteen state in Proposition 3.36 has neither a clean nor a deletion choice, so repeated witness neutralization cannot be assumed to terminate without a new monotone invariant or additional realization information.

Proposition 3.28 (cyclic neutralization dynamics)**.** *On the size-fourteen state of Proposition 3.36, legal applications of Proposition 3.22 contain a directed four-cycle. Consequently no scalar potential can strictly decrease under every legal witness-neutralizing switch. In fact, the exact directed graph on the forty reduced integrable map states is strongly connected, so every potential weakly nonincreasing on all such switches is constant.*

The cycle occurs entirely outside the global-minimum domain of the displayed 18-vertex realization. At one of its states, for every relative circuit translation there is an explicit binary cycle avoiding one affine class and meeting six support edges but only two nonsupport edges.

Proof. Use the notation

$$k = (u_1, u_2, u_3, a, b)$$

from the proof of Proposition 3.36; its forty states have nonzero entries, $a \neq b$, and $b = 3+u_1+u_2+u_3$. Let U interchange 1 and 2 in K. In the following table, Q lists the five component obstruction bits, and T lists $(T_{x,A}, T_{x,B})$ for $x = 0, \dots, 4$:

k	Q	T	S	normalized target
11112	00101	33/11/11/11/22	12	22112
22112	00011	33/22/22/11/22	013	21121
21121	00101	33/22/11/11/11	012	21212
21212	00011	33/22/11/22/22	13	11112

(15n)

In each row the xor of the T_x's selected by S is zero on both circuits. The selected Q-sum is one in the source and zero in the target. Apply U on the named components; if $0 \in S$, follow by the common normalization $U^{-1} = U$. Direct substitution in the five keys gives the displayed target. Thus every arrow is legal, neutralizes its chosen witness, and the fourth returns to the first. This proves the strict-potential assertion without computation.

For the stronger finite statement, the two bundled checkers independently enumerate the forty keys, their four dual witnesses each, and all six map choices. Both obtain 320 witness-switch certificates, 136 distinct directed state pairs, and mutual reachability of all forty states. Paths in both directions force a weakly nonincreasing potential to take equal values everywhere.

Finally take state $k = 21212$. The integrated low word on the second support circuit is

$$\texttt{2313010} + z$$

for an arbitrary relative translation z. In the edge numbering of the displayed realization, the binary cycle

$$C = \{8, 9, 10, 11, 12, 13, 25, 26\} \tag{15o}$$

uses the last six support values $\texttt{313010} + z$, which omit $2 + z$, and two complement edges. Hence $C \cap M_{2+z} = \varnothing$ while

$$|C \cap h| - |C - h| = 6 - 2 = 4.$$

The exchange theorem excludes this state at a global minimum for every relative translation. Exact cycle-space enumeration separately finds a positive exchange gain for all forty states and all four translations, but that larger finite fact is not needed for the displayed descent. □

Remark 3.29. Proposition 3.28 is a no-go theorem for a proof that iterates Proposition 3.22 using only its abstract state. It neither rules out a potential incorporating the four shortest-join inequalities nor weakens the minimum-projection conjecture. It shows that global-minimum data must enter any convergence argument essentially.

Theorem 3.30 (minimum projections through size eleven)**.** *Under the hypotheses of Theorem 3.15, every cardinality-minimum extendable projection of size at most eleven is cleanable.*

Proof. Only total size eleven remains. The four-colour circuit condition gives the three shapes

$$11, \qquad 4 + 7, \qquad 5 + 6. \tag{16}$$

The same exact boundary enumeration used above gives

shape	canonical words	charge-valid	dirty	failed direct repairs
11	345	1,183,710	919,764	0
$4+7$	17	57,500	46,786	0
$5+6$	26	87,884	71,170	14

The 14 literal failures form 11 orbits under the full affine, dihedral, and component-renaming action. For each failure, the checker exhausts the same component maps and independent circuit starts and finds a certificate satisfying (13) in which every low value on the six-circuit is nonzero. Proposition 3.17 then produces an extendable projection supported only on the five-circuit. Its size is five, contradicting the assumed global minimality of the size-eleven projection. Hence every globally minimum size-eleven state is directly cleanable. ☐

Remark 3.31 (sharp direct-repair countermodel). The 14 failed boundary states are realizable. Replacing each two-vertex component block by an edge and each three-vertex block by a claw yields, in all 14 cases, the same simple bridgeless non-Tait graph on 12 vertices, with graph6 encoding

`Kt?G?DIPOqCo`.

It is the triangle-expanded Petersen graph. Exact cycle-space enumeration shows that the displayed size-eleven $5+6$ projection is extendable but uncleanable, while the minimum extendable-projection size is five and all six size-five minima are cleanable. Thus it refutes the direct-repair strengthening, not FiveCDC or the minimum-projection conjecture. The cumulative checker and all 14 certificates are in the bundled through-eleven package.

3.5 Clean or delete and Kempe escape through size fourteen

Propositions 3.14 and 3.17 give a useful dichotomy: it is enough to find component maps and circuit starts that either make every affine cut parity even, or make one whole support circuit low-valued and nowhere zero. The first branch cleans the projection; the second strictly shortens it and is therefore impossible for a global minimum.

Theorem 3.32 (minimum projections through size twelve)**.** *Under the hypotheses of Theorem 3.15, every cardinality-minimum extendable projection of size at most twelve is cleanable.*

Proof. Only total size twelve remains. The complete circuit-length list is

$$12, \qquad 4+8, \qquad 5+7, \qquad 6+6, \qquad 4+4+4. \tag{17}$$

For each shape, the exact classifier enumerates the same word orbits and charge-valid component partitions as before. It first exhausts every component map and relative circuit start for direct cleanliness. If that fails, it tests whether the integrated word on a circuit omits a colour. Translating that circuit by the omitted colour makes all of its low values nonzero, giving the hypothesis of Proposition 3.17. The mutually exclusive counts are

shape	words	dirty	direct clean	delete
12	1014	11,465,978	11,465,978	0
$4+8$	66	773,721	773,717	4
$5+7$	64	738,969	738,763	206
$6+6$	66	774,196	774,014	182
$4+4+4$	3	35,960	35,960	0

There are zero states in neither branch. The 392 deletion states have literal component maps, integrated circuit words, and omitted-colour certificates in the frozen output. Since a globally minimum projection cannot take the deletion branch, every size-twelve minimum is cleanable. ☐

Remark 3.33. The primary C++ classifier and certificate parser are bundled in the through-twelve clean-or-delete package. A separately written C++ exact cover auditor independently regenerates the word orbits, valid component blocks, map search, clean decisions, and all 392 deletion decisions. The two implementations agree on every table entry. This is independent agent replication, not human peer review.

Theorem 3.34 (minimum projections through size thirteen)**.** *Under the hypotheses of Theorem 3.15, every cardinality-minimum extendable projection of size at most thirteen is cleanable.*

Proof. Only total size thirteen remains. Since every circuit component contains all four colours, the complete shape list is

$$13, \qquad 4+9, \qquad 5+8, \qquad 6+7, \qquad 4+4+5. \tag{18}$$

The exact classifier applies the same clean-or-delete test as in Theorem 3.32. Its complete counts are

shape	words	charge-valid	dirty	direct clean	delete
13	2583	155,381,679	129,684,490	129,684,490	0
$4+9$	130	7,753,918	6,657,654	6,657,594	60
$5+8$	203	12,144,485	10,404,652	10,401,428	3224
$6+7$	242	14,480,258	12,415,026	12,410,636	4390
$4+4+5$	4	238,522	208,144	208,144	0

Thus 159,369,966 dirty states split into 159,362,292 direct repairs and 7,674 strict circuit deletions, with no residual state outside those branches. Proposition 3.17 excludes every deletion state from global minimality, proving the claim. ☐

Remark 3.35. The primary size-thirteen package freezes one literal map, integrated word, deleted circuit, and omitted colour for every deletion state. A separately written valid-block/exact-cover implementation regenerates the word orbits, component partitions, map search, affine cleaning decisions, and deletion decisions. The implementations agree on every table entry. This is exact independent agent replication, not peer review.

Proposition 3.36 (sharp failure of unrestricted clean or delete)**.** *The clean-or-delete boundary statement used in Theorems 3.32 and 3.34 is false at total support size fourteen. A counterstate has two seven-circuits with*

$$\begin{aligned} (c_i) &= \mathit{0101023} \mid \mathit{0101232}, \\ (\pi_i) &= \mathit{01234444413024}. \end{aligned} \tag{19}$$

Every circuit uses all four colours and every component charge is zero, but no component-map tuple and relative circuit translation cleans the state or makes either circuit low-valued and nowhere zero.

Proof. The derivative word is `3111121 | 2111311`, from which the five zero block charges and dirtiness are checked directly. Normalize the map of block zero to the identity, and put

$$u_i = L_i(1)\ (i = 1,2,3), \qquad a = L_4(1), \qquad b = L_4(2).$$

All five displayed elements are nonzero and $a \neq b$. The transformed derivatives on the two circuits are

$$(3, u_1, u_2, u_3, a, b, a), \qquad (b, a, u_1, u_3, 3, u_2, a).$$

Their two integrability equations coincide and reduce to

$$b = 3 + u_1 + u_2 + u_3. \tag{20}$$

Zero-start integration gives

$$\begin{aligned} r^A &= (3, 3 + u_1, 3 + u_1 + u_2, b, a + b, a, 0), \\ r^B &= (b, a + b, 3 + u_2 + u_3 + a, 3 + u_2 + a, u_2 + a, a, 0). \end{aligned}$$

Both rows contain $\{0, a, b, a + b\} = K$, so deletion is impossible.

Let z translate the second circuit. Cleanliness at blocks $0, 1, 2, 3$ respectively requires z to lie in

$$\begin{array}{ll} u_2 + a + \{0, 3\}, & 3 + a + b + \{0, u_1\}, \\ 3 + u_1 + u_2 + a + \{0, u_2\}, & u_1 + u_3 + a + \{0, u_3\}. \end{array} \tag{21}$$

If $z = u_2 + a + 3$, the third and fourth lines force $u_1 = u_2$ and $u_3 = 3$; then (20) gives $b = 0$. If $z = u_2 + a$, the second and fourth lines force $u_3 = u_1$ and $u_2 = u_1$; the third then forces $u_1 = 3$, and again (20) gives $b = 0$. Both conclusions contradict invertibility of L_4. Thus no clean translation exists. $\square$

Remark 3.37 (realization and scope). The state (19) is realized by a simple connected bridgeless non-Tait cubic graph on 18 vertices, with canonical graph6 encoding

`Qs???SC@GS@_CDOoC@@@?O?CO?g`.

The graph is nonplanar; the package displays a $K_{3,3}$ subdivision. Exact enumeration of its 2^{10}-element cycle space finds 15,360 ordered extensions of the displayed size-fourteen projection and no clean one. Nevertheless, the graph has minimum extendable-projection size five, exactly four minimum supports, and all four are cleanable. Thus Proposition 3.36 refutes the unrestricted boundary lemma, not Conjecture 4.1 or FiveCDC. The complete certificate and graph audit are in the bundled size-fourteen dichotomy-counterstate package; it also independently replays all 1,296 normalized map tuples and freezes the 40 effective cases. A separately written package exhausts all 6^5 full map assignments and independently reconstructs the graph, cycle space, extension counts, and four minima.

Theorem 3.38 (minimum projections through size fourteen)**.** *Under the hypotheses of Theorem 3.15, every cardinality-minimum extendable projection of size at most fourteen is cleanable.*

Proof. Only total size fourteen remains. Since every circuit component contains all four affine colours, the complete shape list is

$$\begin{array}{c} 14, \quad 4 + 10, \quad 5 + 9, \quad 6 + 8, \quad 7 + 7, \\ 4 + 4 + 6, \quad 4 + 5 + 5. \end{array} \tag{22}$$

The primary exact classifier applies the same component-map, relative-start, cleaning, and deletion tests as above. Its mutually exclusive counts are

shape	words	dirty	direct clean	delete	residual
14	7382	1,748,842,896	1,748,842,896	0	0
$4+10$	416	100,736,602	100,735,166	1436	0
$5+9$	505	121,073,174	121,032,843	40,331	0
$6+8$	809	195,911,662	195,844,374	67,288	0
$7+7$	333	80,104,020	80,066,557	37,239	224
$4+4+6$	22	5,408,156	5,408,156	0	0
$4+5+5$	14	3,401,666	3,401,596	70	0

(23)

The seven rows comprise 9,481 word orbits and 2,616,134,989 charge-valid states. The 2,255,478,176 dirty states split into 2,255,331,588 direct repairs, 146,364 strict circuit deletions, and 224 residuals. The deletion rows contradict global minimality by Proposition 3.17. Every residual has shape $7+7$, with 46 states of component-size profile $6+2+2+2+2$ and 178 of profile $4+4+2+2+2$.

It remains to exclude those 224 states. Fix two distinct nonzero low colours x, y, put $\Delta = x+y$, and consider one component W of $G-h$. Retain the edges of W whose low value is x or y. There is a linear functional $\ell : K \to \mathbb{F}_2$ satisfying

$$\ell(x) = \ell(y) = 1, \qquad \ell(\Delta) = 0. \tag{24}$$

Applying ℓ to the low-flow equation shows that every vertex of W away from h has even retained degree. At the split boundary occurrence i, the unique complement edge has low value d_i. Consequently the odd boundary vertices are exactly the occurrences with $d_i \in \{x, y\}$, and the retained subgraph pairs them by paths.

Interchange x and y on one such path. Each internal flow equation is unchanged, every complement value remains nonzero, and both endpoint derivatives change by Δ. Since both endpoints belong to the same complement component, its boundary charge is unchanged. If the endpoints lie on the same support circuit, its two changes cancel. If they lie on different support circuits, the certificate names another component whose selected two-colour subgraph has exactly two odd boundary occurrences, one on each circuit. Its forced path supplies a second switch by Δ, restoring both circuit closure equations.

The pairing of terminals depends on the unknown realization inside W. For each residual the finite certificate therefore stores a set S of successful endpoint pairs such that

$$M \cap S \neq \varnothing \quad \text{for every perfect matching } M \text{ of the terminals.} \tag{25}$$

There are three possible perfect matchings on four terminals and fifteen on six. For every pair in S, the certificate gives the primary and auxiliary endpoints, all component maps, both integrated circuit words, and a colour omitted from one circuit. Translating by that omitted colour and deleting the circuit produces an extendable projection of size seven.

The exhaustive builder verifies all normalized map choices and produces 724 literal deletion rows. A separately structured checker verifies (25), every path-switch boundary update, every component map and circuit integration, and every omitted-colour deletion. Hence every possible internal path pairing yields a strict deletion. No residual is globally minimum, which proves the theorem. □

Remark 3.39 (scope and independent replication). The path argument uses the split-occurrence boundary model, automatic in a cubic graph: every circuit occurrence is a distinct vertex with one incident complement edge carrying d_i. If several occurrences are identified at one higher-degree vertex, only their xor is visible to the complement two-colour subgraph. Thus the path theorem itself is not a higher-degree boundary theorem. Proposition 1.1 instead replaces the graph by a new cubic expansion and pushes a completed cover back.

A separately written $7+7$ census regenerates all 333 word orbits, 91,481,505 charge-valid states, 80,104,020 dirty states, and the same 224 residual rows using a valid-block exact-cover implementation. Its sorted failure-set digest agrees with the primary census. This is independent agent replication, not independent human peer review. Only the $7+7$ branch has an independently implemented full state census. For the other six shapes—14, $4+10$, $5+9$, $6+8$, $4+4+6$, and $4+5+5$—the word-orbit counts, charge-valid partition counts, and all direct-clean/delete decisions currently rest on the primary classifier.

3.6 A universal tensor repair and the size-fifteen frontier

The one-circuit rows in the preceding finite tables are instances of a general algebraic theorem.

Theorem 3.40 (relative-GL(2, 2) tensor lemma)**.** *Put $\mathcal{G} = \mathrm{GL}(2,2)$. Let V be finite. Orient each unordered pair $\{a,b\} \subseteq V$ once, and assign it an arbitrary linear functional*

$$\phi_{ab} : \mathrm{Mat}_2(\mathbb{F}_2) \longrightarrow \mathbb{F}_2.$$

There are maps $L_a \in \mathcal{G}$ such that, for

$$w_{ab} = \phi_{ab}(L_a^{-1}L_b), \qquad Q_a = \sum_{b\neq a} w_{ab}, \tag{26}$$

every Q_a is zero.

Proof. Regard functions $\mathcal{G} \to \mathbb{F}_2$ as a six-dimensional vector space. Identify $0 = (0,0)$, $1 = (1,0)$, $2 = (0,1)$, and $3 = (1,1)$. Write a map as the string of its images of $0, 1, 2, 3$, and define

$$\lambda(f) = f(\texttt{0132}) + f(\texttt{0213}) + f(\texttt{0321}).$$

Then

$$\lambda(1) = 1, \qquad \lambda(M \mapsto M_{ij}) = 0 \quad (1 \le i,j \le 2). \tag{26a}$$

For the second assertion, in column-major order the flattened matrices of the three displayed maps are

$$(1,0,1,1), \quad (0,1,1,0), \quad (1,1,0,1),$$

whose sum is zero.

For a fixed tensor instance, let

$$F((L_a)_{a\in V}) = \prod_{a\in V}(1+Q_a).$$

This is the indicator of the desired assignments. Apply $\Lambda = \lambda^{\otimes V}$ after expanding F. An expansion choice selects at most one incident edge at each vertex. Because the edge functions are Boolean, $w_e^2 = w_e$, so terms may be grouped by the simple graph H of distinct selected edges.

If a nonempty H has a leaf, its grouped monomial depends on the map at that leaf through one linear coordinate function. Multiplication by a fixed matrix is linear, and on $\mathcal{G}$

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix}^{-1} = \begin{pmatrix} s & q \\ r & p \end{pmatrix}.$$

Thus (26a) kills the monomial, regardless of which orientation was fixed for the leaf edge.

Suppose instead that H has no leaf. If S is the set of vertices that selected an edge, then

$$|E(H)| \le |S| \le |V(H)| \le |E(H)|.$$

The last inequality is the minimum-degree-two bound, with $V(H)$ containing only nonisolated vertices. Equality holds throughout. Consequently H is a disjoint union of cycles, every vertex selects one edge, and every edge is selected exactly once. Each cycle has two such selections, its two cyclic orientations. A nonempty union of cycles therefore has coefficient $2^{c(H)} = 0$ in $\mathbb{F}_2$.

All nonconstant grouped terms are killed or cancel. The constant term gives

$$\Lambda(F) = \lambda(1)^{|V|} = 1.$$

Hence F is not the zero function, proving the theorem. ☐

Corollary 3.41 (universal one-circuit direct cleaning)**.** *Every charge-valid abstract boundary state whose support is one circuit has a direct repair of Proposition 3.14. There is no bound on the circuit length, number of complement components, or component degrees in the abstract state.*

Proof. Use the notation of Proposition 3.14, and assume the single support circuit is cut open in a fixed place. For each component a, conservation says

$$\bigoplus_{\pi_i = a} d_i = 0. \tag{26b}$$

Choose local maps L_a and put $z_i = L_{\pi_i} d_i$. For distinct components a, b, define

$$w_{ab} = \sum_{\substack{j<i \\ \pi_j = b,\ \pi_i = a}} B(z_j, z_i), \tag{26c}$$

where B is the nondegenerate alternating form on $\mathbb{F}_2^2$. The opposite ordering has the same value, since the sum of the two orderings is

$$B\left(\sum_{\pi_j = b} z_j, \sum_{\pi_i = a} z_i\right) = 0$$

by (26b). Thus w_{ab} is symmetric and independent of the cut. For the latter assertion explicitly, moving one occurrence (a, z) across the cut changes its pair bit with b by $B(z, \sum_{\pi_i = b} z_i) = 0$; the case of a b-occurrence is the same. For the orientation fixed on $\{a, b\}$, preservation of B gives

$$B(L_b d_j, L_a d_i) = B(L_a^{-1} L_b d_j, d_i).$$

Summing these terms defines a linear functional of the four matrix entries of $L_a^{-1} L_b$, exactly as in Theorem 3.40. Reversing the chosen pair orientation is harmless because matrix inversion is linear on $\mathcal{G}$.

It remains to identify the obstruction. Integrate the z_i's around the circuit, with an arbitrary base value, to obtain the r_i's in (13). Put $q(x_1, x_2) = x_1 x_2$; its polarization identity is

$$q(x + y) + q(x) + q(y) = B(x, y). \tag{26d}$$

The common parity of the four low colours at component a is the parity of colour 11. At the boundary occurrence i, both support incidences e_{i-1}, e_i must be counted; an edge whose two ends belong to the same component is thereby counted twice and cancels. Hence this parity is

$$P_a = \sum_{\pi_i = a} \big(q(r_{i-1}) + q(r_i)\big).$$

Since $r_i = r_{i-1} + z_i$, (26d) gives

$$P_a = \sum_{\pi_i = a} q(z_i) + \sum_{\pi_i = a} B(r_{i-1}, z_i). \tag{26e}$$

After cutting the circuit, write $r_{i-1} = r_* + \sum_{j<i} z_j$. The base term in (26e) is $B(r_*, \sum_{\pi_i=a} z_i) = 0$ by (26b), so

$$P_a = \sum_{\pi_i = a} q(z_i) + \sum_{\substack{j<i \\ \pi_i = a}} B(z_j, z_i).$$

Separate the last sum according to whether $\pi_j = a$. Applying q to the zero block sum in (26b) gives

$$0 = q\left(\sum_{\pi_i = a} z_i\right) = \sum_{\pi_i = a} q(z_i) + \sum_{\substack{j<i \\ \pi_j = \pi_i = a}} B(z_j, z_i).$$

Thus the $q(z_i)$ terms and all same-block pairs cancel. The remaining cross-component terms are exactly

$$Q_a = \sum_{b \neq a} w_{ab}.$$

Theorem 3.40 chooses the maps so that all these common parity bits vanish. Equations (13) and (14) then hold, so the state directly cleans. □

Corollary 3.42 (circuitwise-balanced several-circuit repair). *Suppose a boundary state has support circuits $D_1, \ldots, D_t$ and satisfies*

$$\bigoplus_{\substack{\pi_i = a \\ i \text{ on } D_j}} d_i = 0 \qquad \textit{for every component } a \textit{ and circuit } D_j. \tag{26f}$$

Then it is directly cleanable.

Proof. Construct (26c) separately on every circuit. For each unordered pair, sum its per-circuit linear functionals. The total obstruction at a component is the sum of its per-circuit obstructions, hence is again the degree parity Q_a of one tensor instance. Apply Theorem 3.40 once to obtain a common map L_a for every circuit. Condition (26f) makes each circuit integrable, since

$$\bigoplus_{i \text{ on } D_j} L_{\pi_i} d_i = \bigoplus_a L_a \left(\bigoplus_{\substack{\pi_i = a \\ i \text{ on } D_j}} d_i \right) = 0.$$

Integrate the circuits independently. □

Remark 3.43. Condition (26f) is stronger than ordinary component conservation, which requires only the xor over all support circuits to vanish. Thus Corollary 3.42 is a structural subclass theorem, not a replacement for the multi-circuit census below.

Proposition 3.44 (Petersen boundary of the tensor repair). *Suppose every component of the complement of a support cycle H has exactly two support-boundary occurrences. Contract the support circuits to vertices and the complement components to edges, obtaining an interaction multigraph J, with a loop when both occurrences lie on the same circuit.*

The feasible images of the component derivatives under independent $\mathrm{GL}(2,2)$ *maps are exactly the nowhere-zero K-flows on J. After integration, either occurrence of an interaction edge a traverses an unoriented edge $\{x, x+t_a\}$ of the K_4 on the four low colours. Component a is clean exactly when its two occurrences traverse the same K_4-edge.*

This subclass is not always directly cleanable. On the Petersen graph, take its two disjoint 5*-circuits as H and its perfect matching as the five complement components. In their cyclic orders the component words are*

$$01234 \mid 03142. \tag{26g}$$

This support has no clean extension. Every extension, however, omits a low colour on exactly one of the two support circuits and therefore admits a strict circuit deletion. In particular, the extension

$$t = 11123, \qquad s|_H = 10130 \mid 13210$$

deletes the first circuit by adding $(1,2)$*, leaving the low word* 32312*. The resulting support has size five, is globally minimum, and is clean.*

Proof. The two derivative occurrences belonging to one complement component have the same nonzero value, by component conservation. Give that common value to the corresponding edge of J. Circuit closure is precisely the flow equation at the corresponding vertex of J, with loops counted twice. Conversely, every nowhere-zero K-flow on J is obtained, because a component map can send its original nonzero derivative to any prescribed nonzero value.

If the integrated value immediately before an occurrence is x, the two adjacent support values are x and $x+t_a$. The parity vector seen by the two occurrences of component a is zero exactly when these two unordered pairs agree. This proves the two dictionaries.

For (26g), J has two vertices joined by five parallel edges. Its flows are the tuples

$$(t_0, \dots, t_4) \in \{1,2,3\}^5, \qquad t_0 + \cdots + t_4 = 0. \tag{26h}$$

Their colour multiplicities have pattern $(3,1,1)$, so there are $3 \cdot 5!/(3!1!1!) = 60$. The global $\mathrm{GL}(2,2) \cong S_3$ action is free, and the ten orbit representatives have the following numbers of integrated colours on the two circuits:

t	11123	11213	11231	12113	12131	12223	12232	12311	12322	12333
$\#A/\#B$	3/4	4/3	3/4	4/3	4/3	3/4	4/3	3/4	4/3	3/4.

Each entry follows by five successive xors in the two orders (26g). If an extension were clean, the two circuits would traverse the same five assigned K_4-edges. Their vertex sets would both be the union of the endpoints of those edges and would have equal cardinality, contradicting the table. The same table shows that every flow deletes exactly one circuit.

The displayed words integrate the first row. After its stated deletion, the second 5-circuit is the support. Petersen is non-Tait-colourable and has girth five, so no extendable projection has size below five. The complement of the remaining circuit is connected, hence its only component has empty cut and the projection is clean. □

Remark 3.45 (exact minimality within the subclass)*.* Two independently structured enumerators check every loopless two-occurrence multi-circuit boundary of total size four, six, and eight; every flowable state is cleanable. Thus (26g) is the smallest two-occurrence abstract counterstate to direct cleaning. The full orbit table, literal Petersen realization, descended flow, exact census, two checkers, and hash ledger are in `scratch/minimum-projection-two-occurrence-interaction-20260729/`. This is a boundary-method obstruction, not a minimum-projection or FiveCDC counterexample.

Theorem 3.46 (loopless two-occurrence frontier through eighteen)**.** *Suppose every complement component has exactly two support-boundary occurrences, the interaction multigraph J of Proposition 3.44 is loopless, the total support is at most eighteen, and J admits a nowhere-zero K-flow. Then some such flow either cleans the state or omits a low colour on a whole support circuit and hence strictly deletes that circuit. Consequently every cardinality-minimum extendable projection in this subclass is cleanable.*

Proof. The total support is the degree sum of J. Disconnected interaction components may be treated independently. A vertex of degree at most three already deletes under every feasible flow, because its integrated closed walk visits at most three of the four points of K. If every degree is even, the constant nonzero flow makes every local walk alternate between two points and deletes.

A connected two-vertex interaction graph is a parallel bundle. For an even bundle, use the constant flow. For an odd bundle of size at least three, give two consecutive edges distinct values b, c and every other edge $a = b+c$. Both vertex xors vanish, while the local walk starting before the exceptional pair has prefixes $0, b, a$ and then alternates between a and 0, omitting c. A connected three-vertex graph which is not a triangle is a path of parallel bundles; apply the same zero-xor construction independently at its two leaves.

It remains to consider a connected noneulerian graph of minimum degree four on at least three vertices. Five vertices would have degree sum at least twenty, so J has at most four vertices.

For three vertices, write the positive triangle multiplicities as (a, b, c). After the preceding degree and Eulerian reductions, the complete finite list through degree sum eighteen is

$$\begin{gathered}(3,2,2),\ (3,3,2),\ (4,3,1),\\ (5,2,2),\ (4,4,1),\ (4,3,2).\end{gathered} \tag{26i}$$

Indeed, the other positive partitions of at most nine edges have a degree-two or degree-three vertex or have all three degrees even.

For four vertices the only remaining degree multiset is $(5, 5, 4, 4)$. Writing the multiplicities in the order $(01, 02, 03, 12, 13, 23)$, enumeration of the nonnegative solutions modulo all 24 vertex permutations gives nine orbits. One is disconnected, one has a bridge and therefore no nowhere-zero flow, and the seven finite cases are

$$\begin{gathered}(0,1,3,3,1,1),\ (0,1,3,3,2,0),\ (0,1,3,4,1,0),\\ (0,2,2,2,2,1),\ (0,2,2,2,3,0),\\ (1,1,2,2,1,2),\ (1,1,2,3,1,1).\end{gathered} \tag{26j}$$

A standard-library enumerator checks the complete 90-row labelled degree solution and this orbit reduction.

For the thirteen profiles in (26i)–(26j), the exact census is

$$
\begin{array}{c|r|r|r}
\text{profile} & \text{cyclic-order states} & \text{clean} & \text{delete only} \\
\hline
T(3,2,2) & 432 & 432 & 0 \\
T(3,3,2) & 8{,}640 & 8{,}640 & 0 \\
T(4,3,1) & 12{,}960 & 12{,}744 & 216 \\
T(5,2,2) & 388{,}800 & 388{,}800 & 0 \\
T(4,4,1) & 362{,}880 & 356{,}256 & 6{,}624 \\
T(4,3,2) & 259{,}200 & 259{,}200 & 0 \\
Q(0,1,3,3,1,1) & 1{,}296 & 1{,}296 & 0 \\
Q(0,1,3,3,2,0) & 1{,}296 & 1{,}296 & 0 \\
Q(0,1,3,4,1,0) & 1{,}296 & 1{,}188 & 108 \\
Q(0,2,2,2,2,1) & 1{,}296 & 1{,}296 & 0 \\
Q(0,2,2,2,3,0) & 1{,}296 & 1{,}296 & 0 \\
Q(1,1,2,2,1,2) & 1{,}296 & 1{,}296 & 0 \\
Q(1,1,2,3,1,1) & 1{,}296 & 1{,}296 & 0 \\
\hline
\text{total} & 1{,}041{,}984 & 1{,}035{,}036 & 6{,}948
\end{array}
\tag{26k}
$$

The primary checker enumerates every distinctly labelled-edge cyclic order modulo rotation and reversal and every nowhere-zero K-flow. It integrates the local walks, exhausts their relative translations, and tests the alternating-form cleaning equations and deletion. A separately written checker constructs the literal two-element K_4 endpoint sets instead of using that equation. The two implementations reproduce every row of (26k); there are no residual states.

Finally, by the interaction dictionary, a deleting flow is realized by component maps. Toggling the omitted low colour on that circuit removes the entire circuit and yields a strictly smaller extendable projection, which is impossible at a cardinality minimum. □

Remark 3.47. The complete replay, literal-endpoint independent checker, frozen outputs, and hash ledger are in `scratch/two-occurrence-clean-delete-through18-20260729/`. Graph looplessness does not prevent two occurrences of one complement component on the same support circuit. The following theorem closes that case through support sixteen.

Theorem 3.48 (two-occurrence loops through sixteen)**.** *Suppose every complement component has exactly two support-boundary occurrences, the total support is at most sixteen, and the interaction multigraph J of Proposition 3.44 admits a nowhere-zero K-flow. Loops in J are allowed. Then some feasible flow cleans the state or strictly deletes a support circuit. Consequently every cardinality-minimum extendable projection in this subclass is cleanable.*

Proof. If J is loopless, apply Theorem 3.46; assume it has a loop. Delete the loops but retain all vertices, and work componentwise in the resulting core. On a one-vertex component, giving every loop one fixed nonzero value makes its local walk alternate between two points and deletes. We may therefore assume the core is connected on at least two vertices. Since every interaction edge has two occurrences, the total support is even. If a local walk omits a colour it deletes, so every surviving support circuit has length at least four. There are consequently at most four interaction vertices.

First record the local loop move. If the two occurrences of a loop have value $t \neq 0$ at derivative positions $i < j$, change its component map so that their common value becomes $s \neq 0, t$. With $\Delta = t + s$, the integrated support word changes by

$$c'_k = c_k + \Delta \quad (i \le k < j), \qquad c'_k = c_k \quad \text{otherwise.} \tag{26l}$$

This is a legal graph move: in the loop component the t, s-subgraph has exactly the two displayed odd boundary vertices, whose joining path may be switched. Up to affine colour and dihedral circuit symmetry, the only dirty words of lengths four and five are `0123` and `01023`. Their equal-derivative pairs are respectively two and three pairs; direct substitution in (26l) gives a word using at most three colours in every case. Thus a loop on a circuit of length at most five deletes.

If all circuit lengths are even, put one fixed nonzero value on every interaction edge, loops included. Every vertex xor vanishes and every local walk alternates between two points, so it deletes. Four vertices therefore force lengths $4+4+4+4$ and are done.

For two vertices the loopless core is a parallel bundle. In the only remaining odd case, the shorter circuit has length five or seven. At length five it has five ports and no surviving loop. At length seven it has $3, 5$, or 7 ports; one port is a bridge and is not flowable. For the other cases, a finite paired-port check chooses zero-xor port values and arbitrary nonzero loop values. Modulo rotation and reversal, the exact pattern counts for $(\text{length}, \text{ports})$ are

$$(5,3) : 10, \quad (5,5) : 1, \quad (7,3) : 105, \quad (7,5) : 21, \quad (7,7) : 1. \tag{26m}$$

Every one of these 138 patterns has a flow whose local walk omits a colour.

For three vertices, the only length triples surviving the preceding reductions are $4+5+7$ and $5+5+6$. Every loop lies on the unique long circuit. Removing one or two loops leaves, up to the exchange of equal-length vertices, precisely the four core-multiplicity profiles shown below. More loops leave a bridge or a separate one-circuit component. If δ_v denotes the loopless-core degree at v, the multiplicities follow from

$$2m_{01} = \delta_0 + \delta_1 - \delta_2, \quad 2m_{02} = \delta_0 + \delta_2 - \delta_1, \quad 2m_{12} = \delta_1 + \delta_2 - \delta_0.$$

The exact literal-endpoint census is

lengths	(m_{01}, m_{02}, m_{12})	loops	states	clean	delete only	residual
$4+5+7$	$(2,2,3)$	1	6,480	0	6,480	0
$4+5+7$	$(3,1,2)$	2	3,240	0	3,240	0
$5+5+6$	$(3,2,2)$	1	4,320	168	4,152	0
$5+5+6$	$(4,1,1)$	2	2,304	12	2,292	0
total			16,344	180	16,164	0

(26n)

The checker distinctly labels parallel edges, enumerates cyclic orders modulo rotation and reversal and all normalized nowhere-zero K-flows, integrates every local walk, exhausts relative translations, and compares the two literal K_4-edges of every interaction edge. The preceding one-, two-, three-, and four-vertex reduction is exhaustive, proving the claim. As in Theorem 3.46, deletion contradicts cardinality minimality. □

Remark 3.49. The human reduction, exact verifier, first local failures at length six, and hash ledger are in `scratch/support16-loop-higheroccurrence-reduction-20260729/`. This combines with Theorem 3.46 to cover all two-occurrence states through support sixteen and all loopless such states through support eighteen. Higher occurrence multiplicities are not covered.

Theorem 3.50 (minimum projections through size fifteen)**.** *Under the hypotheses of Theorem 3.15, every cardinality-minimum extendable projection of size at most fifteen is cleanable.*

Proof. Only total size fifteen remains. The complete circuit-length list is

$$\begin{gathered} 15, \quad 4+11, \quad 5+10, \quad 6+9, \quad 7+8, \\ 4+4+7, \quad 4+5+6, \quad 5+5+5. \end{gathered}$$

The primary exact classifier applies the same canonical word generation, charge-valid set-partition enumeration, component maps, relative circuit starts, direct-cleaning test, and strict-deletion test as above. Its mutually exclusive counts are:

shape	words	charge-valid	dirty	direct clean	delete	residual
$4+4+7$	34	45,252,514	40,901,504	40,901,444	60	0
$4+5+6$	72	94,943,832	85,691,028	85,687,790	3,238	0
$5+5+5$	20	26,498,996	23,865,564	23,865,124	440	0
$4+11$	985	1,305,883,765	1,166,954,048	1,166,933,276	20,772	0
$5+10$	1,489	1,975,366,285	1,762,347,048	1,761,806,487	540,561	0
$6+9$	1,924	2,555,260,324	2,281,097,772	2,280,299,726	798,046	0
$7+8$	1,964	2,609,251,412	2,328,991,036	2,327,987,350	997,650	6,036
15	20,004	26,634,745,428	23,398,774,592	23,398,774,592	0	0
total	26,492	35,247,202,556	31,088,622,592	31,086,255,789	2,360,767	6,036

Corollary 3.41 proves the 15 row without enumeration. As an exact regression, sixteen completed anchor/Laplacian shards also test all 23,398,774,592 dirty one-circuit states and find zero failures. A separately implemented zero-sum-partition recurrence reproduces the one-circuit charge-valid, initially clean, and dirty counts.

For the other seven shapes, every deletion row contradicts global minimality by Proposition 3.17. Only the 6,036 $7+8$ residual rows remain. They are handled by a finite, realization-independent lift from the size-fourteen Kempe certificates. Delete one removable repeated-colour occurrence whose two adjacent derivative occurrences lie in the same complement block. If the old derivatives are a, b, the shorter boundary replaces them by $a + b$; charge conservation is unchanged and no two blocks are merged.

For a fixed two-colour Kempe pair, terminal status is a linear functional of the derivative. Every perfect matching of the old terminals collapses to one on the shorter boundary. If both a, b are terminals and are matched to p, q, the collapsed matching has the artificial pair pq; switching its two actual old paths has the same boundary action as the short switch, apart from an explicitly tracked change of the reinserted support value. The other cases merely delete the pair ab, rename one terminal as $a + b$, or leave the matching unchanged. Thus a matching-robust short certificate lifts after finitely checking the reinserted value, circuit closure, and final omitted-colour deletion.

A separately written canonical induction checker verifies every one of the 6,036 emitted residual rows. Each has between three and five same-block smoothing positions, and at least one of them smooths canonically to a frozen size-fourteen residual. Thus each row is canonically equivalent to a member of the 6,180-state inverse-extension family. A literal checker quantifies over every old perfect matching and verifies a successful one- or two-path lift for all 6,180 states; a redundant direct checker verifies the same realization-robust game on all 6,036 emitted rows. Hence every cubic realization of every size-fifteen residual has a strict deletion escape. No residual can be globally minimum, proving the theorem. □

Remark 3.51 (size-fifteen trust boundary). The tensor proof is human-checkable and independent of the one-circuit census. The zero-sum recurrence independently checks its partition totals; the anchor parser checks shard completeness and zero failures but imports the primary word/partition generator. The multi-circuit word-orbit counts, charge-valid partition counts, and clean/delete decisions rest on the primary classifier. The residual induction checker is independently written, but takes the 6,036 primary residual rows as input. It finds 5,200 fully canonical residual classes, with digest

`f2c0c2b53322c72f00672e3f9c7d807aeb26a6ba771357a673c84c2fdaf96206`.

All source, shard outputs, literal residuals, frozen totals, and hashes are in the bundled size-fifteen exact-frontier, induction-audit, anchor-audit, and one-circuit tensor packages. These computational claims are exact agent-produced certificates, not independent human peer review.

3.7 One fixed-word support-sixteen matching frontier

We next isolate one exactly completed subcase at the first unrestricted support size. This uses the standard multipole viewpoint, in which a three-edge-colouring induces a parity-constrained state on the semiedges [3], and the usual two-colour Kempe switch [13]. The result below concerns one canonical support-word orbit and one round of one-colour-pair path switches only.

Fix two support 8-circuits with low word and boundary derivative word

$$c = \texttt{01010123} \mid \texttt{01012302}, \qquad d = \texttt{31111131} \mid \texttt{21113132}. \tag{26o}$$

For a pair $P = \{x, y\} \subseteq K - \{0\}$, put $\Delta = x + y$. In a complement component W, the x, y-coloured subgraph has odd boundary set

$$T_W(P) = \{i \in \partial W : d_i \in P\}. \tag{26p}$$

Its path components pair the terminals; write this perfect matching as $M_W(P)$. Nature may supply any abstract perfect matching on every set $T_W(P)$. After observing them, one may switch a nonempty subset S of the displayed paths, all for the same pair P, and then choose arbitrary feasible component maps and relative circuit translations. Call S charge-closed when it contains an even number of paths whose endpoints lie on different support circuits. The precise quantifier order of the matching-observed game is

$$\boxed{\begin{array}{c} \exists P \;\; \forall \big(M_W(P)\big)_W \\ \exists\, \varnothing \neq S \subseteq \dot{\bigcup_W} M_W(P) : \\ S \text{ is charge-closed and the new state cleans or deletes.} \end{array}} \tag{26q}$$

Only matchings for the single chosen P occur in one play.

Proposition 3.52 (fixed-word support-sixteen matching frontier)**.** *For the one word orbit represented by* (26o)*, the exact canonical component-partition census has*

outcome	*states*
directly clean	8,041,808
strict circuit deletion only	4,514
neither direct outcome	8
charge-valid total	8,046,330.

(26r)

The eight residual component words are

$$\begin{array}{ll} \mathit{0001234000314200} & \mathit{0001234003144422} \\ \mathit{0012344041300022} & \mathit{0012344044130244} \\ \mathit{0012344400314200} & \mathit{0012344403144422} \\ \mathit{0123444441300022} & \mathit{0123444444130244}. \end{array} \tag{26s}$$

Six of them satisfy (26q) *with the fixed pair* $P = \{1, 3\}$*. The remaining two are precisely*

$$\pi_A = \mathit{0001234000314200}, \qquad \pi_B = \mathit{0123444444130244}, \tag{26t}$$

both of profile $8+2+2+2+2$. For each of the three pairs $\{1,2\},\{1,3\},\{2,3\}$, each state has a jointly graph-realizable adverse matching under which no nonempty charge-closed one-round path subset cleans or deletes. Each adverse triple has a simple connected bridgeless cubic realization on 42 vertices and 63 edges. Both realizations are Tait-colourable, so their actual minimum projection is empty.

Proof. The fixed-word classifier canonically generates every charge-valid set partition of the sixteen derivative occurrences in (26o), exhausts independent $\mathrm{GL}(2,2)$ maps on its blocks and all relative circuit translations, and tests direct cleaning and strict deletion. The word orbit has 512 normalized images and trivial stabilizer. The mutually exclusive totals are (26r). A separately written evaluator replays the eight rows (26s); every row has exactly 320 feasible map tuples and zero clean and zero deletion tuples.

Switching a displayed path adds Δ at its two endpoints. A path with both endpoints on one support circuit preserves both circuit charges; a cross-circuit path changes both charges by Δ. Consequently a path subset is legal exactly when it contains an even number of cross-circuit paths, proving the charge-closed test in (26q). The exact matching enumeration gives:

residual profile	number of residuals	$\{1,3\}$ matchings rescued
$5+4+3+2+2$	2	9/9
$6+3+3+2+2$	2	15/15
$6+4+2+2+2$	2	9/9
$8+2+2+2+2$	2	14/15.

(26u)

For the first six rows the numerator equals the denominator for the single fixed pair $\{1,3\}$, so (26q) holds even when Nature is allowed every abstract matching, including matchings not supplied by a particular graph realization.

For either row in (26t), the rescued/total counts for $\{1,2\},\{1,3\},\{2,3\}$ are respectively

$$14/15, \qquad 14/15, \qquad 2/3. \tag{26v}$$

The unique adverse systems have respectively $6,7,3$ paths and $31,63,3$ nonempty charge-closed subsets. Literal replay of all 97 subsets leaves exactly 320 feasible map tuples and no clean or deletion tuple after every switch.

It remains to show that the three adverse matchings can occur simultaneously. Use a properly three-edge-coloured K_4-e two-pole for every two-occurrence component. For the eight-occurrence component take ten vertices $0,\ldots,9$ and the following internal colour matchings:

$$\begin{array}{r|l} 1 & 24,\ 67,\ 89 \\ 2 & 06,\ 13,\ 28,\ 59 \\ 3 & 03,\ 19,\ 48,\ 57. \end{array} \tag{26w}$$

Vertices $0,\ldots,7$ carry the eight semiedges, whose colours are $1,1,3,1,2,1,3,2$. Attach them to support occurrences in the orders

$$\begin{array}{r|l} \pi_A & 2,1,0,7,8,9,14,15 \\ \pi_B & 4,5,6,7,8,9,14,15. \end{array} \tag{26x}$$

Following the three bichromatic path systems in (26w) gives exactly the unique adverse matching in (26v) for every colour pair. Joining these multipoles to the two support circuits gives the two asserted 42-vertex graphs. A literal graph audit checks distinct edges, degree three, connectedness, bridgelessness, the full flow equations, all three induced matching systems, and all 97 one-round subsets. It also supplies a Tait colouring of each graph, proving that the zero projection is extendable and minimum. $\square$

Remark 3.53 (scope and verification boundary). Proposition 3.52 completes one canonical $8+8$ word orbit, not the other $8+8$ word orbits or the other support shapes of size sixteen. Its adverse graphs have nonminimum displayed projections. The result excludes only one support-preserving round using one colour pair; it says nothing against a second Kempe round, a support-changing exchange, or additional restrictions forced by global minimality. In particular it is neither a FiveCDC counterexample nor a general minimum-projection theorem at size sixteen.

The complete census, residual file, human reduction, two checkers, literal graph encodings, Tait colourings, and hash ledgers are in the two packages

`scratch/unrestricted-support16-interaction-frontier-20260729/`
`scratch/support16-residual-fourterminal-matching-game-20260729/`.

A clean-room agent audit, which imports neither candidate checker, is in

`scratch/support16-residual-fourterminal-matching-game-blind-audit-20260729/`.

The matching certificate has the following SHA-256, split across two lines:

`96a5ff4201c411c278e03ea9941a0665`
`669575bb07da73aeff59d0cca38c6966`.

Corollary 3.54 (rainbow shortcut). *Every binary cycle C satisfying $|C \cap h| > |C - h|$ meets all four sets M_c.*

Proof. If C omitted one class M_c, it would contradict (6). □

4 The full-flow problem and failure of minimum selection

Theorem 3.1 suggests the following selection principle.

Conjecture 4.1 (minimum extendable projection; false). Every bridgeless cubic graph has a minimum-cardinality extendable projection that is cleanable.

This statement, together with the Hušek–Šámal equivalence and Proposition 1.1, would imply the standard FiveCDC conjecture for every finite bridgeless multigraph. It survives all finite tests through support fifteen, but Theorem 4.3 below refutes it. Because the constructed minimum projection is unique, the stronger assertion that every minimum projection is cleanable fails at the same time.

Proposition 4.2 (full-flow master exchange). *Let $\mu(G)$ be the minimum cardinality of an extendable projection and, for a low flow $s' : E \to K$, let $Z(s') = \{e : s'(e) = 0\}$. Then*

$$\mu(G) = \min_{\substack{s' \text{ a } K\text{-flow} \\ J \subseteq E - Z(s') \\ \partial J = \partial Z(s')}} \big(|Z(s')| + |J|\big),$$

where ∂A denotes the set of odd-degree vertices of an edge set A. Every feasible $Z(s')$ is a matching. For a feasible triple $M = Z(s')$, $J_1 = J$, and $h' = M \,\dot\cup\, J_1$, the extension (h', s') is clean if and only if there is a second ∂M-join

$$J_2 \subseteq E - (M \cup J_1).$$

Proof. If (h', s') is nowhere zero and $M = Z(s')$, then $M \subseteq h'$. Put $J = h' - M$. Since h' is a binary cycle, the disjoint edge sets satisfy $\partial J = \partial M$. Conversely, given s', M, and such a J, the edge set $h' = M \dot\cup J$ is a binary cycle. The flow (h', s') is nowhere zero because the low coordinate vanishes exactly on M, where the first coordinate is one. These constructions are inverse and preserve the displayed objective.

At a cubic vertex, a low flow has zero on either zero, one, or all three incident edges. The last case is impossible in a feasible triple: ∂M contains the vertex, while the disjoint J has degree zero there. Hence M is a matching. For fixed M, the inner minimization is the standard shortest-∂M-join problem in $G - M$.

It remains to prove the last assertion. Let W be a component of $G - h'$, and let n_c be the parity of the number of cut edges with low value $c \in K$. Low-flow conservation gives $n_1 = n_2 = n_3$. Since h' is a binary cycle and $\delta(W) \subseteq h'$, cut parity gives $n_0 + n_1 + n_2 + n_3 = 0$; hence all four parities are equal. Moreover

$$n_0 = |M \cap \delta(W)| = |\partial M \cap W| \pmod 2.$$

Thus the extension is clean precisely when every component of $G - h'$ contains an even number of ∂M-vertices. By the elementary T-join existence criterion, this is equivalent to a ∂M-join contained in $G - h'$, which is the asserted J_2. □

In this form Conjecture 4.1 says exactly that some optimum master triple admits the second join. For a fixed extension (h, s), the four branches in Corollary 3.2 are only the low flows $s' = s + ch$, $c \in K$. They do not exhaust the master minimum. In the 278-vertex graph of Proposition 6.3, all four of those branches have cost fourteen, while a coupled low flow has

$$M = \{1, 5\}, \qquad J = \{0, 2, 3, 4, 6\},$$

in the frozen base-edge order and therefore cost seven. Its difference from the displayed flow uses five nonzero Fano values, so it is not a constant-vector circuit switch. The complete proof and two independently structured literal-graph checkers are in `scratch/minimum-projection-full-flow-exchange-20260729/`.

Theorem 4.3 (strict-lock counterexample to minimum selection)**.** *There is a simple connected bridgeless nonplanar cubic graph G with*

$$|V(G)| = 162, \qquad |E(G)| = 243$$

whose unique minimum-cardinality extendable projection H has size 54, is the disjoint union of two 27-circuits, and is not cleanable. Nevertheless G has a five-cycle double cover.

Proof. Start with the audited 18-vertex base B from Proposition 6.3 before inflation. In its frozen edge order the target $h = \{0, \ldots, 13\}$ is the union of two 7-circuits. Exact enumeration of the 2^{10}-element cycle space finds 1,008 extendable projections. The target has 15,360 ordered low-flow extensions and none is clean.

Let S be the simple bridgeless cubic graph with canonical graph6 encoding

```
Q???C@?K?WEOM?_aGo?I__W@G_?
```

and distinguish edge $e_* = 5 = (5, 10)$ in the frozen order. If $a_b(e_*)$ is the minimum projection size of a nowhere-zero $\mathbb{F}_2^3$-flow on S, conditional on the first bit of e_* being b, exact cycle-space enumeration gives

$$a_0(e_*) = 7, \qquad a_1(e_*) = 5.$$

The a_1-minimum support is unique and is the 5-circuit $\{5, 13, 14, 16, 17\}$. Delete e_* and attach one external link at each exposed endpoint. The resulting two-pole has effective costs

$$w_0 = 7, \qquad w_1 = 6.$$

Indeed the odd case removes the projected edge e_* and adds two projected links. An explicit first-coordinate-preserving shear sends the displayed odd terminal value to any prescribed odd terminal value, so the same support attains the bound in every odd terminal class.

For $L \subseteq h$, replace precisely the base edges in L by fresh copies of this two-pole. If k is another extendable terminal projection of B, its cost relative to the target is

$$\Delta_L(k) = |L \cap (h - k)| - |(h - L) \cap (h - k)| + |k - h|.$$

Testing all 1,008 extendable k against all 2^{14} sets L shows that eight locks are necessary and sufficient; there are 180 minimum placements. For

$$L = \{0, 2, 3, 4, 7, 8, 9, 10\}$$

the minimum value of $\Delta_L(k)$ over $k \neq h$ is one.

The substituted graph has $18 + 8 \cdot 18 = 162$ vertices and $27 + 8 \cdot 27 = 243$ edges. Direct checks establish simplicity, cubicity, connectivity, and bridgelessness; the base is a nonplanar minor. Contracting any nowhere-zero flow through the eight poles gives a nowhere-zero flow on B, so the local costs and the displayed Δ_L bound every projection below by

$$14 + 5 \cdot 8 = 54.$$

The target lift attains this value. Equality forces terminal support h, and the unique local odd minimum then forces a unique literal projection H, consisting of two 27-circuits.

Every extension of H contracts to an extension of h. The old components of $B - h$ persist, and the two terminal links replacing a base support edge carry its contracted value. Hence cleanliness upstairs would imply cleanliness of the contracted base extension, contradicting the 15,360-extension calculation. Thus H is uncleanable.

Finally, the graph has a human-checkable five-cover. Fix two-subset edge labels giving five-cycle double covers of B and S. For each substitution, permute the five labels on its copy of S so that the label on e_* equals the label on the replaced base edge. Delete both old edges and give both terminal links that common two-subset label. At every exposed vertex a like-labelled incidence is replaced by a like-labelled incidence, so all five parities remain even; every new edge still has exactly two labels. The resulting 243-label certificate and a separately generated satisfying assignment of the mandatory exact-two/XOR formula are both checked literally. $\square$

The lock placement and both graph constructions were independently reimplemented. They canonicalize to the identical graph6 row. The primary package, complete edge list, five-cover certificates, and ledger are in `scratch/minimum-projection-strict-parity-lock-20260729/`; the separately written audit is in the sibling strict-lock independent audit package. This theorem closes the proposed minimum-selection route negatively. FiveCDC must select a nonminimum clean projection on this graph. Theorem 3.50 remains informative: no such counterexample can have minimum support at most fifteen, whereas this witness has minimum support 54.

5 Exact finite census

5.1 Canonical order-18 method

The accompanying checker uses canonical generation rather than sampling. For order 18, the command

```
geng -cq -d3 -D3 18
```

generates every connected simple cubic graph once. The checker then:

1. tests bridgelessness by deleting every edge and checking connectivity;
2. tests Tait colourability by exhaustive backtracking;
3. enumerates the entire binary cycle space;
4. processes nonzero projections h by increasing cardinality;
5. quotients pairs of cycles (p, q) by their exact semantic state $(p \cup q, p \cap q)$;
6. tests extendability by $E - h \subseteq p \cup q$; and
7. computes the components of $G - h$ and tests (5) on every component.

For a connected cubic graph of order 18, the binary cycle-space dimension is

$$|E| - |V| + 1 = 27 - 18 + 1 = 10.$$

Thus all $2^{10} = 1024$ binary cycles are enumerated. Equal union/intersection pairs have exactly the same coverage and component parity behaviour, so the semantic quotient loses no information. No snark or cyclic-connectivity reduction is used.

5.2 Results and reproducibility

class	count
connected simple cubic graphs	41,301
bridgeless graphs	39,866
bridgeless non-Tait graphs	179

Table 1: Complete order-18 population.

Every minimum-cardinality extendable projection of every one of the 179 non-Tait graphs is cleanable. The minimum-size histogram is

$$5 : 166, \qquad 6 : 11, \qquad 7 : 1, \qquad 8 : 1.$$

The checker hashes, in canonical graph order, the graph6 string, minimum size, number of minimum extendable projections, and number of cleanable minimum projections. The resulting SHA-256 is

`82b01cc2f01d4f50f7745f4bdb1b3dc46ade798d452a9d4f11d4a91d738a7834`.

From the project root, run:

```
python3 scratch/check_husek_samal_minimum_projection_frontier.py \
  --order 18
```

The checker requires Python, the sibling cycle-space module `scratch/search_fano_all_bad_projection_subspaces.py`, and nauty `geng` on `PATH`. It has no SAT solver, network, or non-standard Python-package dependency. The relevant source hashes are

frontier checker:
`37e972782863c7a9506f6d339159ddaca3865363107704f86365c1a9b77a30c3`
cycle-space module:
`7e5ea2599e7533f041ea88de5c6f5c7a3bf542cedc33d78728439d43ce8861e8`

This finite theorem explains why minimum selection was initially plausible, but Theorem 4.3 disproves the universal statement beyond the tested range.

5.3 Expanded frozen-source replay

A separate C++ classifier implements the same extendability and cleanability semantics by Gaussian elimination over $\mathbb{F}_2$. It replays two previously frozen graph6 populations:

1. 14,009 cyclically 4-edge-connected non-Tait records of even orders 10 through 28; and
2. 12,892 bridgeless non-Tait rows from a broader hard order-22 source.

The exact results are:

frozen source	graphs	minimum projections	uncleanable
cyclically 4, orders 10–28	14,009	92,754	0
hard order 22	12,892	54,785	0
total	26,901	147,539	0

Table 2: Exact minimum-extendable-projection replay on frozen sources.

For the cyclically 4 source the minimum-size profile is

$$5 : 14007, \qquad 6 : 1, \qquad 9 : 1.$$

For the hard order-22 source it is

$$5 : 11776, \quad 6 : 1025, \quad 7 : 55, \quad 8 : 20, \quad 9 : 6, \quad 10 : 10.$$

These counts refer to graphs; the third column of the table counts all minimum projections on those graphs.

From the project root, run:

```
cd scratch/minimum-projection-census-through28-20260729
python3 verify_summary.py
```

The replay compiles the classifier, verifies every input digest and every emitted graph6 row, and checks all aggregates above. Its checksum ledger is the `SHA256SUMS` file in that package. The unconditional claim here concerns the literal frozen files. Their identification with complete graph classes uses the documented upstream canonical-generation and filtering packages; this replay does not independently regenerate those populations.

5.4 Retained strong snarks of orders 34 and 40

A separate exact scan covers the literal retained strong-snark files at orders 34 and 40: Every row

order	graph6 rows	minimum projections	uncleanable
34	7	371	0
40	7,654	433,359	0
total	7,661	433,730	0

Table 3: Exact minimum-projection scan on the frozen strong-snark rows.

has minimum extendable-projection size ten. An ordinary-CNF solver encodes the two low binary cycles, coverage on $E - h$, product variables for $p_e \wedge q_e$, and an XOR chain imposing even product parity on every component cut. Every even projection through weight twelve is enumerated as a vertex-disjoint union of circuits, which is exhaustive for a cubic graph. All rows reach their first extendable level at weight ten.

The order-34 output and the first five order-40 rows were reproduced independently by the linear-algebra scanner used in the earlier census. The full order-40 claim uses the exact SAT implementation. The concatenated result digest is

`becb7e4b648c000ec874509bccdf389df24f6f0d82bce2da49ba10d5b10d38e3`.

Inputs, frozen output, both scanner implementations, and the replay script are under `scratch/`, in the subdirectory `minimum-projection-known-strong-snarks-20260729/`. As before, the unconditional statement concerns the literal bundled rows; completeness of the named graph populations is inherited from upstream provenance.

5.5 A certified 130-vertex parameter-separation test

The retained 130-vertex exchange graph gives a substantially larger exact stress test. A three-coordinate CNF uses binary-cycle variables (h_e, p_e, q_e), the three parity equations at every vertex, the nowhere-zero clauses $h_e \vee p_e \vee q_e$, and a sequential cardinality counter. Its certified results are

$$\min |h| = 42, \qquad \#\{h : |h| = 42,\ h \text{ extendable}\} = 11264,$$

and every one of those 11,264 supports is cleanable. The size-$\leq$ 41 CNF has 7,760 variables and 16,218 clauses. A second CNF blocks every listed size-42 support and proves enumeration completeness. Both UNSAT LRATs are accepted by C `lrat-check` and verified CakeML `cake_lpr`; every positive clean lift is also checked directly.

On the same graph the different parameter

$$\rho_3(G) = \min_{f,\ a\neq 0} |f^{-1}(a)|$$

equals five, while no size-five value class packs two boundary joins. Thus minimizing one nonzero Fano value class and minimizing an entire coordinate projection are not interchangeable. The former selection principle fails on this graph; the latter passes all 11,264 minimum cases. To replay the package, run:

```
cd search/minimum-projection-n130-20260729
python3 verify.py
```

6 Discussion

The exchange theorem converts global support minimality into four ordinary geodesic conditions: one shortest join for each affine value class. Theorems 3.9, 3.12, 3.15, 3.30, 3.32, and 3.34, together with Theorems 3.38 and 3.50 close every possible nonzero support size through fifteen. Thus Theorem 4.3, whose minimum projection has size 54, is consistent with the sharp proven lower frontier: no counterexample to the false selection principle has minimum projection size at most fifteen. The useful feature is simultaneity. A cycle that would shorten the projection is allowed to be blocked by one colour, but Corollary 3.54 forces it to be blocked by all four. Likewise, disconnected projections are severely constrained because every circuit component must contain every colour. The size-fourteen state (19) shows that this numerical frontier is sharp for the support-preserving clean-or-delete boundary method. Theorem 3.38 also shows exactly what that abstract state forgets: two-colour subgraphs inside a cubic complement pair the boundary occurrences by paths, and a matching-robust Kempe switch reaches a strict deletion. Corollary 3.41 removes one entire unbounded branch: every one-circuit boundary directly cleans. The remaining open frontier therefore begins with several circuits lacking the separate per-circuit balance of Corollary 3.42; larger such states need not admit the same finite strategy. Theorems 3.46 and 3.48 remove the entire two-occurrence branch through size sixteen and the loopless branch through size eighteen. This does not move the unrestricted frontier beyond size fifteen: a surviving state at size sixteen may have a component with more than two occurrences. Proposition 3.52 examines exactly one such fixed-word frontier. A matching-observed $\{1, 3\}$ path choice removes six of its eight direct residuals, but two literal $8 + 2 + 2 + 2 + 2$ systems defeat every one-round choice for every colour pair. Because both realizations are Tait-colourable, this is a clean separation between the one-round boundary game and the globally minimum problem, not an extension of Theorem 3.50.

Proposition 3.3 rules out the most direct LP repair of this gap. The rainbow quotient shore itself carries zero coefficient in every optimal shortest-join dual, before any witness-neutralizing switch. The positive information is forced into cuts that split complement components. This both yields Corollary 3.4 and identifies the missing geometric object: a descent proof must transport interior distance, blossom, or Kempe-pairing information across a neutral recolouring. Proposition 3.5 completely describes the zero-slack boundary of this argument. There the complement is a forest and all colour loads and affine witnesses are exactly balanced; an uncleanable case cannot occur below support twenty-four. What remains is therefore either this rigid large equality regime or positive dual slack.

Proposition 3.24 supplies the first dynamic coupling to global minimality. A concentrated inside/outside derivative colour profile forces an explicit strict descent after one legal common map switch. The remaining implication is exact but open: rainbow-oddness is a parity condition, whereas (15l) is a load inequality that can be shielded by colour-balanced interior vertices. Proposition 3.25 realizes that shielding already on the cube, so local rainbow forcing is false. Because its projection is not globally minimum, a proof must either extract a load violation from genuinely global information or exploit the interior path geometry directly.

Proposition 6.1 (static exchange has no further local boundary force)**.** *Let G be a finite connected simple bridgeless cubic graph, let H be a 2-regular binary cycle, and let $f = (h, s)$ be a nowhere-zero $\mathbb{F}_2^3$-flow with* $\operatorname{supp}(h) = H$. *Suppose every circuit component of H contains every value of $s|_H : E(H) \to \mathbb{F}_2^2$. There are a simple connected bridgeless cubic graph G' and a nowhere-zero flow $f' = (h', s')$ on G', with* $\operatorname{supp}(h') = H$ *and $s'|_H = s|_H$, carrying the same support circuits, complement-component partition, and two-colour boundary path pairings, for which*

$$C \cap M_c = \varnothing \quad \Longrightarrow \quad |C \cap H| \le |C - H| \tag{27}$$

for every $c \in \mathbb{F}_2^2$ *and every binary cycle* C *of* G'*, where* $M_c = \{e \in H : s'(e) = c\}$*.*

Proof. Let $e = uv \notin H$ have nonzero low colour $t = s(e)$, and let $\{t, x, y\} = \mathbb{F}_2^2 - \{0\}$. Delete e, add vertices a, b, p, q, and add

$$ua, bv, pq \text{ of colour } t, \qquad ap, bq \text{ of colour } x, \qquad aq, bp \text{ of colour } y. \tag{28}$$

Give every new edge first coordinate zero and retain f on every unreplaced edge. Every new vertex sees the three nonzero low colours, so both flow equations are preserved and the resulting flow is nowhere zero. For either pair $\{t, x\}$ or $\{t, y\}$, the bichromatic path joins the old endpoints u, v; the $\{x, y\}$ component is internal. Hence the substitution preserves every original two-colour boundary pairing. Every u-to-v traversal through the two-pole, including the attachment edges, has length at least four. Chaining k copies gives old-endpoint distance $L = 3k+1$, while preserving simplicity, cubicity, connectedness, and bridgelessness.

Apply the same chain length to every edge of $G - H$, choosing $L \geq |H|$. Decompose a binary cycle avoiding M_c into edge-disjoint circuits. A circuit contained in one replacement chain uses no support edge. Every other circuit either traverses a chain from one old endpoint to the other and hence uses at least L non-support edges, or is a circuit component of H. The latter is impossible because every component of H meets M_c. Thus each remaining circuit D satisfies

$$|D - H| \geq L \geq |H| \geq |D \cap H|.$$

Summing over the circuit decomposition proves (27). □

Remark 6.2 (a one-round obstruction, not a minimum counterexample)*.* The bundled literal instance has 230 vertices, 345 edges, and support equal to two 8-circuits. Two checkers establish all four families (27): one exhausts 2^{22} base binary cycles and the other independently solves the four shortest-T-join problems in the inflated graph. The affine-class sizes are $(6, 5, 3, 2)$, the four join sizes are $(10, 11, 13, 14)$, and the four minimum containing-cycle sizes are 16. All 97 nonempty charge-restoring one-round fixed-colour path multiswitches fail to clean or delete even after every componentwise affine repair. Nevertheless, switching the $\{1, 2\}$-paths with boundary endpoints 1–10 and 4–15, then recomputing and switching the $\{1, 3\}$-path $5-6$, makes colour 2 absent from the first support circuit and permits its deletion. The graph is Tait-colourable, so its actual minimum projection is empty. The example therefore refutes neither Conjecture 4.1 nor FiveCDC. It shows that the force of global minimality must be used dynamically after each support-neutral recolouring. The proof and replay are in `scratch/general-kempe-descent-static-exchange-obstruction-20260729/`.

Proposition 6.3 (a goal-free survivor of every static filter)**.** *There is a simple connected bridgeless cubic graph on 278 vertices with an uncleanable extendable projection* h *of size fourteen such that:*

1. *h has a rainbow-odd six-map-integrable witness shore, but none of its six common switches cleans or deletes a support circuit;*
2. *all nine colour-load inequalities hold; and*
3. *for every* $c \in K$*, the shortest* T_c*-join in* $G - M_c$ *has size* $|h| - |M_c|$*.*

Nevertheless the graph's exact minimum extendable-projection size is seven. Thus the conjunction of all listed static consequences of global minimality does not imply cleaning or deletion.

Proof. Begin with two support 7-circuits, low words and component word

$$r = \texttt{3231320} \mid \texttt{1320320},$$
$$\pi = \texttt{01234444413024}.$$

Choose the union of component blocks $0, 1, 3$. The derivative word is `3112212` | `1212312`; the selected derivatives on each circuit are one copy each of $1, 2, 3$, so every common linear map is integrable. Direct integration of the six maps gives obstruction bits

$$(1, 1, 0, 1, 0, 1),$$

and every resulting support circuit still uses all four colours. Hence none of the six choices cleans or deletes. The selected derivative counts are $(2, 2, 2)$, the outside counts are $(4, 4, 0)$, and the 18-vertex base realization has four off-support vertices, so the largest load is $6 \leq 8$.

Replace each of the thirteen base complement edges by a chain of five two-poles from the proof of Proposition 6.1. The old-endpoint distance is sixteen, larger than $|h| = 14$; therefore all four complete exchange families hold. Exact shortest-join calculation sharpens this to

c	0	1	2	3
$\lvert M_c\rvert$	3	2	4	5
shortest T_c-join	11	12	10	9
containing cycle	14	14	14	14.

Every low flow with fixed projection h contracts through the two-pole chains to a base low flow with the same component cut parities. The exact 1,024-cycle base enumeration has 15,360 extensions of h and no clean one, so the inflated projection is uncleanable.

For an arbitrary full flow, contraction gives a nowhere-zero base flow. An old support edge with first coordinate one costs one inflated support edge, while a contracted complement edge with first coordinate one forces both terminal links of its chain into the inflated support and costs at least two. Exhausting the 1,024 base binary cycles and the 66,207 simultaneously missed sets gives weighted minimum seven. Conversely an explicit base flow supported on the first 7-circuit is even on every replaced complement edge and lifts through all chains with support still seven. This proves the exact global minimum. ☐

Remark 6.4. The through-thirteen boundary theorem makes support fourteen smallest for a state with no clean or deletion component-map tuple. The full six-map table, 18- and 278-vertex constructions, two exact implementations, hostile audit, outputs, and ledger are in the rainbow-load-survivor package under `scratch/`. The example is a proof-method obstruction, not a globally minimum projection or a FiveCDC counterexample.

Proposition 6.5 (a directly clean unbounded class)**.** *Let $H = \operatorname{supp}(h)$ be one circuit. If every component W of $(V, E - H)$ contains exactly two vertices of H, then every extension $f = (h, s)$ is cleanable by independent componentwise* $\mathrm{GL}(2, 2)$ *maps followed by reintegration of the low values on H.*

Proof. At a vertex v_i of H, let $d_i \neq 0$ be the low value on the unique edge of $E - H$, and let s_{i-1}, s_i be the low values on the two incident edges of H. Flow conservation gives

$$d_i = s_{i-1} + s_i. \tag{29}$$

Summing the low-flow equations over a complement component W shows that the xor of its boundary values d_i is zero. There are exactly two, so they are equal. Fix $t \in \mathbb{F}_2^2 - \{0\}$. Independently on each W, choose a linear automorphism sending its common boundary value to t. These maps preserve the low-flow equations inside W.

The circuit H has two vertices for every complement component and hence even length. Thus (29), now with every $d_i = t$, integrates around H, and its edge values alternate between a and $a+t$. At each of the two vertices of H in W, the two support incidences therefore have one value a and one value $a + t$. Across the four incidences both values occur twice. A support edge internal to W is counted twice; deleting its two equal copies does not change parity. Consequently every low value occurs evenly on $\delta(W)$. This holds for every W, so the resulting extension is clean. □

Proposition 6.6 (conditional orbit-reflecting inflation)**.** *Let a support-boundary state realized in a simple connected bridgeless cubic graph have no clean or circuit-deletion state in one connected component of its reconfiguration graph, where the moves include all charge-preserving two-colour Kempe path multiswitches and all feasible componentwise* $\mathrm{GL}(2, 2)$ *maps. There is a simple connected bridgeless cubic inflation for which:*

1. *every reachable boundary-changing move contracts to a move in the same base component, while every additional internal move contracts to the identity; and*

2. *at every reachable state, all four inequalities*

$$C \cap M_c = \varnothing \quad \Longrightarrow \quad |C \cap H| \leq |C - H| \tag{30}$$

   hold for every binary cycle C and every $c \in \mathbb{F}_2^2$.

The assertion is conditional: it does not assert that such a goal-free base component exists or that the displayed support is globally minimum.

Proof. Delete an edge ab of $K_{3,3}$, regard a, b as degree-two terminals, and attach one external edge at each terminal. Call this two-pole P. By the parity lemma, the attachment colours in every proper three-edge-colouring of P are equal, say to t; adding ab in colour t gives a Tait colouring of $K_{3,3}$, and deletion is the inverse. The two Tait-colouring classes of $K_{3,3}$ have every colour pair Hamiltonian [13, Section 4.2]. Therefore a pair containing t induces the unique terminal-to-terminal path of P, while the other pair induces an internal circuit. A switch of the first kind changes the boundary exactly as a switch on the contracted edge; a switch of the second kind leaves the boundary fixed. Both retain the two-valued internal Kempe-class label. Hence P preserves and reflects boundary-changing Kempe moves.

Replace every complement edge by a chain of r copies of P. The shortest terminal-to-terminal path inside one copy has length three, so the old-endpoint distance of the chain is $L = 4r + 1$. Equal terminal colours propagate along a chain. Thus every boundary-changing Kempe component contracts to its base component, every internal switch contracts to the identity, and every base move expands. Contraction also preserves the endpoints of simultaneous path switches and the boundary action of componentwise maps. Fresh pole vertices preserve simplicity and cubicity. Bridgelessness follows because a base circuit through a replaced edge expands through a terminal path, while every internal pole edge lies on an internal bichromatic circuit or on a terminal path completed by such a base circuit.

Choose $L \geq |H|$. In any reachable state, absence of a deletion means that every circuit component of H uses all four low values. Fix c and decompose a binary cycle C avoiding M_c into edge-disjoint circuits. A circuit internal to a replacement chain meets no support edge. Every other circuit using complement edges traverses a whole chain, so it has at least L complement edges and at most $|H|$ support edges. A circuit contained in H is impossible because every support component meets M_c. Summing over the decomposition gives (30). Orbit reflection prevents a new clean or deletion boundary state, and the same argument applies after every reachable recolouring. □

Remark 6.7 (exact replay and search boundary). The bundled dynamic Kempe frontier package contains a dependency-free enumeration of all twelve proper pole colourings, its two six-state Kempe orbits, chain distances through eight copies, and all 146,599 two-terminal pairings on one support circuit through order fourteen. A separately written checker scans all 3^9 edge assignments of $K_{3,3}$ and verifies the cut-incidence parity in Proposition 6.5. A noncanonical exploratory search over 75 larger random poles found no goal-free component; this is negative experimental evidence, not an exhaustive theorem.

The four-colour simultaneity cannot be replaced by a theorem about one matching. In the Petersen graph with edge set

$$\{03, 06, 09, 14, 16, 18, 25, 26, 27, 37, 38, 47, 49, 58, 59\},$$

the matching $M = \{06, 14, 25\}$ is contained in exactly two minimum-cardinality binary cycles:

$$\{06, 09, 14, 18, 25, 26, 49, 58\}, \qquad \{06, 09, 14, 16, 25, 27, 47, 59\}.$$

Both have size eight, and in both cases the components of the complement have M-endpoint parities $1, 0, 1$. Exhausting all 2^{15} edge subsets proves minimality. The complete finite replay is in `scratch/petersen-minimum-tjoin-countermodel-20260729/`. This obstruction does not realize four synchronized affine classes.

Theorem 4.3 supplies the exact counterexample to Conjecture 4.1: the positive interior-shore information and even full global minimality do not force cleaning. The surviving FiveCDC problem is therefore not to repair a minimum projection universally, but to select a possibly nonminimum clean projection. No theorem in this note carries out that selection for every bridgeless graph. In particular, the exact support-sixteen fixed-word census and matching game of Proposition 3.52 do not change the resolution status: they cover one word orbit, and their two adverse examples have empty minimum projection.

AI-use and verification disclosure

This research route, proof draft, checker, computation, and exposition were developed by OpenAI Codex agents under Atharva Vaidya's direction. The minimum-projection exchange proof, the size-at-most-five proof, and the span proof used at sizes six and seven are displayed in full so that they can be checked without trusting an AI system. Those boundary classifications and the stronger direct-cleaning theorem through size ten are also regenerated by dependency-free exact checkers. The through-eleven checker freezes every first direct-repair failure and its smaller-support certificate. The through-twelve package freezes all 392 new circuit-deletion certificates and includes a separately implemented full census. The matching through-thirteen packages independently reproduce all five new shapes and freeze all 7,674 deletion certificates. The size-fourteen primary package freezes all seven shape counts and the 224 residual rows. A separately written full $7+7$ enumerator regenerates all 333 word orbits and 91,481,505 charge-valid states, while the Kempe package freezes 724 matching-robust deletion rows and checks them with an independently structured literal verifier. Codex agents also derived the universal tensor lemma, the one-circuit obstruction identity, and the circuitwise-balanced corollary displayed above. A separate hostile agent audit checked the expansion grouping, the leaf cancellation, all no-leaf graphs through seven tensor vertices, and all 4,096 three-vertex tensors; this remains agent review, not independent human verification. Codex agents also derived the fixed-map quadratic reduction and the six-map witness-neutralization identity. A separately written hostile checker recomputed the 16-entry local table and verified both identities on 126,258 balanced circuit rows through length eight. This is an independent implementation by another agent, not independent human review. Codex agents also found the exact neutralization cycle and wrote its two dynamics checkers. The four-arrow cycle and the gain-four exchange descent are displayed for direct human checking; the forty-state strong-connectivity and 160-profile exchange census remain computational claims. Codex agents also derived the shortest-join zero-price and support-density theorems. Their universal claims rest on the displayed LP-duality proof; two separately written finite audits check the six-map recomputation algebra but are not a substitute for human review. A hostile Codex audit separately checked the primal/dual formulation, the every-optimal-dual quantifier, the density argument, the recomputation laws, and the package hash ledger. The exact density-slack decomposition and equality classification were derived by Codex agents and are displayed in Proposition 3.5. Two arithmetic audits and a hostile logical audit checked the identity, forest inference, colour and topology counts, witness balance, size corollary, and hash ledger. The six-map colour-load inequality and its constructive descent were also derived by Codex agents. Its complete functional proof is displayed in Proposition 3.24; two independently structured finite audits check the map/count identities and boundary conversion, and a hostile Codex audit checked every logical step and the frozen ledger. Codex agents also found the cube counterstate in Proposition 3.25. A separate hostile audit reconstructed its canonical graph, flow, complement stars, six-map table, rainbow and load calculations, local minimality, Tait colouring, two checkers, and hash ledger. Codex agents also derived the two-occurrence interaction dictionary and Petersen descent. A separate hostile audit checked the complete ten-orbit table, the analytic 60-flow and 320-map counts, the literal Petersen realization, the globally minimum clean descendant, both smaller-size enumerators, and the package ledger. This is likewise agent review rather than independent human verification. Codex agents also proved Theorem 3.46; two independently structured exact checkers reproduce its 1,041,984-state partition, including the 1,019,952 new support-eighteen states. A separate agent audit checked the graph-shape reduction, semantics, counts, implementation independence, and hash ledger. This is agent review, not independent human verification. Codex agents also proved Theorem 3.48; its human reduction and exact checker cover the 16,344 finite loop-profile states, and a separately written blind audit reconstructs the marked-word orbits, shape list, literal endpoint tests, and ledger. This too is agent review, not independent human verification. The size-fifteen primary package freezes all eight shape counts and 6,036 residual rows. A separately written induction checker canonicalizes every residual and links it to the matching-robust inverse-extension family. The single-circuit anchor census completed all sixteen shards and found zero failures; an independent parser checks aggregation and agreement with the zero-sum-partition recurrence, but it is not an independently implemented full candidate census. Codex agents also generated the 8,046,330-state fixed-word support-sixteen census, formulated the matching-observed game in Proposition 3.52, generated its complete certificate, found both terminal reattachments, and wrote the literal graph audits. The primary census and separately written residual evaluator agree on all eight residuals. The matching checker and graph audit pass with the frozen digest displayed above. A clean-room agent audit independently reconstructs the eight states, all matching quantifiers and counts, both 42-vertex graphs, their induced path systems, and their Tait colourings without importing either candidate checker. These are exact agent-produced computations and a human-readable finite reduction, not independent human peer review. The static-exchange package contains the human proof of Proposition 6.1, a primary full-cycle checker, a separate shortest-T-join checker for the four inequalities, and the complete boundary-profile Kempe graph for its example. Codex agents also combined that inflation with the size-fourteen counterstate to obtain Proposition 6.3. Two exact implementations and two hostile audits checked the abstract map census, 18- and 278-vertex graphs, all four joins, contraction uncleanability, weighted lower bound, explicit size-seven lift, corrected replay instructions, frozen outputs, and hash ledger. Codex agents also derived the full-flow master formulation in Proposition 4.2. One hostile audit supplied the omitted cut-parity derivation and extended the independent checker to reconstruct the full 278-vertex graph before publication. The strict parity lock and the 162-vertex graph in Theorem 4.3 were then found and minimized by Codex agents. Primary and independently written packages enumerate the two 18-vertex cycle spaces and all 2^{14} lock placements, construct the same canonical 162-vertex graph, prove the unique size-54 minimum and contraction obstruction, and check multiple literal FiveCDC certificates. A third hostile implementation separately reconstructed the graph, minimum, dirty-extension counts, and a satisfying FiveCDC assignment. An initial exploratory solver call used a malformed three-input XOR API and was discarded; no datum from that call appears in the frozen certificates. These are exact agent-produced computations and human-readable reductions, not independent human peer review. Separate packages also check the first failure of the unrestricted dichotomy, its 40-case map certificate, and its exhaustive 18-vertex realization analysis. The standard graph-class reduction and its D_5-label checker were likewise written by Codex agents; a separate hostile agent audited the structural proof, which is still not independent human review. The order-18 result is reproducible from canonical generation and exhaustive cycle-space enumeration. The expanded census exactly replays frozen sources but inherits their population provenance. Both remain computational results and should be independently rerun. The static-exchange proposition and example were found, formalized, and hostilely audited by Codex agents. That audit corrected the distinction between the gadget's internal-terminal distance and its old-endpoint traversal distance; it is still not independent human review. The two-terminal cleaning proof and the orbit-reflecting $K_{3,3} - e$ inflation were likewise proposed, formalized, checked, and hostilely audited by Codex agents. The dynamic package contains two independently written finite checkers, but its unbounded conclusions rest on the displayed human proofs rather than the finite tests. The $K_{3,3}$ Kempe-class fact is prior work of Belcastro and Haas.

Hušek and Šámal's flow criterion and Hoffmann-Ostenhof's matching/join formulation are prior work. So are the Xie–Zhang flow-pair formulation, Edmonds–Johnson T-join duality, Fano-flow terminology, the multipole parity framework, and ordinary edge-Kempe switching. This draft makes no claim of a FiveCDC resolution and no priority claim for the minimum-projection viewpoint pending expert literature review. Before public submission, a human graph theorist should check the proof line by line, rerun the census in a clean environment, audit the bibliography, and decide authorship and venue policy under the applicable AI-disclosure rules.

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